The Pulickal Notebook

Volume III

Recursive Structures, Geometric Certificates, Finite Representations, and Epistemic Computation

Pearl Bipin Pulickal

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Preface: What This Notebook Is Trying to Do

This volume develops a collection of mathematical investigations that repeatedly move between abstract structure, finite representation, computation, and questions about physical or epistemic realization. Its purpose is not to force these subjects into one theory, but to determine what can be stated precisely within each model and where the mathematics naturally stops.

The recurring discipline is simple. A definition is not a theorem; a theorem about a mathematical model is not automatically an engineering result; and an engineering or physical claim requires its own observation model and evidence. The chapters therefore separate exact statements from conjectures, research programmes, and experiments. The aim is a finite form of closure: every chapter should end either in a proof, in a standard theorem with its hypotheses made explicit, or in a clearly identified question that remains open.

The volume is intentionally self-contained. Historical formulations are retained where they illuminate the mathematical question, but they are not treated as evidence for stronger conclusions than the mathematics supports. The present edition also incorporates two previously standalone preprints: a branch-sensitive analysis of root limits and iterated attractors, and a Homotopy Type Theory treatment of equality, congruence, similarity, and quotient shape spaces for Euclidean triangles. They are integrated here as mathematical programmes rather than appended as independent papers.

Abstract

This volume reconstructs and develops a family of mathematical programmes spanning branch-sensitive complex-root dynamics and limiting behaviour, recursive-coordinate structures, geometric exclusion certificates, equality and quotient constructions for Euclidean triangle shape, finite-resolution representations, approximation of infinite mathematical structures, operational depth, epistemic refinement, physical distinguishability, and contextual computation.

The first mathematical core studies root limits and recursive complex roots with explicit attention to domains and principal-branch conventions. It begins with the high-order root limit $x^{1/n} \to 1$, continues through signed real subsequences, dimensional interpretation, complex principal roots, Hausdorff convergence of the full root sets to the unit circle, and the exact principal-branch iteration $z \mapsto \mathrm{i}/\sqrt{z}$. It then separates those pointwise dynamics from the distinct formal-coordinate construction of recursively generated radicals. Exact radial and angular dynamics are derived, including the attracting two-cycle of the principal branch. The associated formal recursive-coordinate spaces are treated as finite-dimensional coordinate spaces with an explicit realization map into the complex plane, making the resulting redundancy and quotient structure precise.

A second geometric core develops an exact certificate for points lying outside an anisotropic ellipsoid, followed by a quotient-based account of Euclidean triangle equality and shape. The latter separates strict equality of located triangles from congruence and similarity, constructs the corresponding set quotients, and realizes similarity classes concretely by normalized side lengths. The result is a sound one-sided exclusion certificate and clarifies why a certificate is not automatically a complete anomaly classifier.

The volume then replaces volume-based counting of real numbers by finite-state and metric notions of representation. Covering numbers, finite codebooks, operational resolution, compactness, and error propagation provide a mathematically controlled language for finite approximation. This framework is used again in the treatment of infinite mathematical structures: finite observable nets and controlled discretization are separated from stronger claims about physical Hilbert-space dimension.

A related programme connects geometric models with finite Markov decision processes. The relevant ingredients are stated explicitly: compact operational sectors, finite approximations, action discretization, approximate Markov closure, bounded rewards, and discounted dynamic programming. Classical Q-learning convergence is kept separate from the geometric approximation problem.

The epistemic component studies finite Kripke structures through iterative behavioral partition refinement. The refinement process is shown to stabilize after finitely many steps and, at stabilization, to yield a bisimulation quotient. A proposed universal logarithmic decay law is rejected in favor of a decision-relative notion of epistemic gain.

The final sections examine operational categorical depth and context-dependent computation. Total boundedness yields a finite observational saturation theorem under explicit hypotheses, while the so-called God Monad is recast as the standard Reader or environment monad. Across

all programmes, the guiding objective is the same: replace ambiguous global claims by exact objects, explicit assumptions, finite proofs, and testable boundaries.

Status and Reading Guide

The status words used throughout the notebook are literal.

- **Theorem:** proved under the stated assumptions.
- **Proposition:** a proved local structural consequence.
- **Historical formulation:** an original idea recorded for context without being endorsed.
- **Corrected formulation:** a mathematically valid replacement for a stronger statement that fails.
- **Research programme:** a precisely stated question whose resolution requires further work.
- **Experiment:** a claim that requires data, implementation, or physical measurement.

A repeated methodological distinction is

$$\text{definition} \not\Rightarrow \text{theorem} \not\Rightarrow \text{engineering performance} \not\Rightarrow \text{physical law}.$$

The central chapters move from formal recursive structures and geometric certificates through finite representation and approximation, then to epistemic refinement, physical distinguishability, and contextual computation. The appendices record detailed derivations and verification arguments so that the strongest statements can be checked independently of the narrative development.

Audit Method

For each programme, the analysis follows the same finite protocol: identify the primitive objects and their domains; verify that every operation is defined on the stated domain; distinguish object equality from isomorphism, approximation, or analogy; check limits, extrema, probabilities, and information counts against their hypotheses; attempt explicit counterexamples to stronger claims; prove the surviving statement by a finite chain of lemmas or by a cited standard theorem; and finally ask whether any algorithmic, physical, or empirical conclusion requires an additional modelling assumption. The audit stops once the remaining gap is either a stated hypothesis, a standard theorem with its assumptions, or a genuinely open problem.

> Mathematical boundary
>
> **Terminology.** An infinity-category is not the same object as an infinite-dimensional vector space, Hilbert space, or other dimension-theoretic construction. The word infinity refers to categorical levels of morphism data. Any claim about physical realizability therefore requires an explicit observation or encoding model.

Contents

Chapter 1

Adversarial Audit: What Survives and What Fails

1.1 Executive audit

The previous foundation was substantially correct in its negative conclusions, but several statements can be made stronger. In particular, the recursive-root programme admits a precise quotient-algebra statement; the ellipsoid result admits an exact arbitrary-dimensional theorem; the finite-approximation programme admits a compactness theorem; the epistemic programme admits a bisimulation-stabilization theorem; and the categorical-saturation conjecture can be replaced, at the mathematical level, by a total-boundedness theorem. The physical connection remains conditional because an entropy bound is not by itself a theorem about categorical truncation.

Attack target	**Failure mode**	**Repair**
Recursive imaginary dimension	Linear dependence in $\mathbb{C}$	Treat $\mathcal{H}_N$ as a formal coordinate space and prove its complex realization has rank at most two.
Recursive imaginary algebra	Multiplication by realization is not automatically well-defined	Use the quotient by $\ker \rho_N$; any compatible quotient is at most two-dimensional over $\mathbb{R}$.
Principal-root dynamics	Wrong attracting fixed point / branch blindness	Use the piecewise argument map and prove convergence to a two-cycle.
Root limits and branch conventions	Domain/branch ambiguity	Separate positive, zero, negative, principal-complex, multivalued, and iterated-root problems before taking limits.
HoTT triangle equality	Quotient equality conflated with raw path constructors	Use proposition-valued equivalence relations and set quotients/effectiveness rather than untruncated path generators.
Ellipsoid anomaly rule	$2a_j$ fails for short axes in anisotropic ellipsoids	Replace it by the exact piecewise formula for M_j.
Spherical "cage" interpretation	One-sided exclusion mistaken for complete classification	Define a certificate C and explicitly identify the unresolved region.
Practical number count	Volume equated with cardinality	Replace cardinality by codebook size, covering number, and metric entropy.

Attack target	**Failure mode**	**Repair**
Physical finitization	Entropy bound equated with finite Hilbert-space dimension	State a conditional operational distinguishability theorem instead.
Hitchin $\to$ finite MDP	Missing compactness, discretization, and stack-level observables	Insert an explicit finite-net and error-propagation layer before reinforcement learning.
Q-learning claim	Classical convergence theorem applied to an undefined MDP	Separate MDP convergence from geometric approximation.
Operational Depth	Real-valued arithmetic silently used as discrete iteration	Restrict discrete depth to orbit-compatible inputs and treat continuous depth separately.
Epistemic decay law	Logarithmic law derived without sufficient data	Replace with exact partition refinement, saturation, and decision-relative utility.
Epistemic "$N = 2$" security claim	Model-specific observation threshold treated as universal	Compute depth under an explicitly specified protocol model.
Categorical depth	Abstract categorical height conflated with observable structure	Define level-indexed probe sets and observational saturation.
Resource-limited saturation	Previously left as a bare conjecture	Prove finite δ-saturation from total boundedness of the observable union.
God Monad philosophy	Expressiveness used as evidence for theology	Retain the Reader monad as exact mathematics and separate philosophical preference as a non-theorem.

Audit conclusion

The strongest general lesson is not that every original claim was false. It is that several promising ideas become mathematically useful once the hidden intermediate object is inserted: a quotient, a finite net, a metric observable, a hitting-time target, a refinement partition, or an explicit operational probe family.

Chapter 2

Global Mathematical Method

2.1 The recurring problem

The nine source programmes repeatedly move between four logically distinct levels:

1. a definition of a new object;
2. a theorem about that object;
3. a statement about an algorithm or physical implementation;
4. a philosophical or empirical conclusion drawn from the preceding levels.

A definition does not entail a theorem, a theorem does not entail an engineering speedup, and a mathematical model does not by itself establish an empirical law. Notebook III will retain these distinctions throughout.

2.2 Convergence is not one concept

The present corpus contains several distinct limiting mechanisms. A recursive complex sequence may converge pointwise or by subsequences; a geometric approximation may converge in Hausdorff distance; a probability law may converge weakly; a numerical discretization may converge as the mesh is refined; an epistemic partition may stabilize after finitely many refinements; and an iterative optimization algorithm may converge in probability. These are different statements and will never be collapsed into the phrase "the sequence converges."

2.3 A unifying abstraction

A useful common language is repeated application of an operator

$$X_{n+1} = F_n(X_n), \tag{2.1}$$

followed by a question about the resulting trajectory. The question can be:

$$\begin{array}{ll} \text{dynamics:} & X_n \to X_\infty, \\ \text{hitting:} & \tau_A(x) = \inf\{n : F^n(x) \in A\}, \\ \text{stabilization:} & \Pi_n = \Pi_{n+1}, \\ \text{approximation:} & d(X_n, X) \to 0, \\ \text{optimization:} & Q_n \to Q^*. \end{array} \tag{2.2}$$

Several of the source programmes become substantially clearer when written in this language, including the root-limit, recursive-root, hitting-time, epistemic, approximation, and quotient programmes. A second recurring rule is that a limiting statement is inseparable from its domain and representation: principal and multivalued roots, strict equality and quotient equality, and exact and finite-resolution models are different constructions.

Chapter 3

Approaching Unity: Principal Roots, Branches, and an Iterated Attractor

3.1 The real root sequence

3.1.1 Positive inputs

Fix $x > 0$ and define

$$a_n(x) = x^{1/n} = \exp\left(\frac{\ln x}{n}\right).$$

Theorem 3.1 (Convergence to unity)**.** *For every fixed $x > 0$,*

$$\lim_{n\to\infty} x^{1/n} = 1.$$

Proof. Since $\ln x$ is fixed and $1/n \to 0$, we have $\ln x/n \to 0$. Continuity of the exponential gives

$$x^{1/n} = \exp(\ln x/n) \to \exp(0) = 1.$$

$\square$

Proposition 3.2 (Exact monotonicity)**.** *If $x > 1$, then $a_n(x)$ is strictly decreasing to 1. If $0 < x < 1$, then $a_n(x)$ is strictly increasing to 1. If $x = 1$, the sequence is constant.*

Proof. The sign of $\ln x$ determines the sign of the exponent. For $x > 1$, $(\ln x)/n$ is positive and strictly decreases with n; for $0 < x < 1$, it is negative and strictly increases toward zero. The exponential is strictly increasing. $\square$

Proposition 3.3 (First-order asymptotics)**.** *For fixed $x > 0$,*

$$x^{1/n} = 1 + \frac{\ln x}{n} + O\left(\frac{1}{n^2}\right).$$

In particular,

$$n\bigl(x^{1/n} - 1\bigr) \longrightarrow \ln x.$$

Proof. Put $u_n = \ln x/n$. Then $u_n = O(1/n)$ and $e^{u_n} = 1 + u_n + O(u_n^2)$. $\square$

3.1.2 Zero and negative inputs

The condition $x > 0$ is not cosmetic. At $x = 0$, one has $0^{1/n} = 0$ for every positive integer n, so the limit is 0, not 1.

For $x < 0$, the expression does not define a real number for even n. For odd n, there is a unique real root and

$$x^{1/n} = -|x|^{1/n}.$$

Thus, along odd indices,

$$\lim_{m\to\infty} x^{1/(2m+1)} = -1.$$

There is therefore no single real sequence indexed by all positive integers to which one can assign the limit -1; rather, the odd-indexed subsequence has that limit.

Remark 3.4. The sign behavior for negative inputs is branch-dependent once the problem is moved to the complex plane. In particular, the principal complex root of a negative real number approaches 1 as the root order tends to infinity, not -1. The real odd-root subsequence and the complex principal branch answer different questions.

3.2 Dimensional Interpretation and Nondimensionalization

3.2.1 What fractional powers do to units

A physical quantity is not merely a real number. It consists of a numerical value together with a unit or dimension. The SI framework permits derived quantities to carry products of powers of base dimensions, so taking a fractional power of a dimensional quantity is not, by itself, a logical contradiction. For example,

$$\sqrt{L}$$

has the dimension $L^{1/2}$. The correct question is whether such a derived quantity is an observable or otherwise meaningful quantity in the physical model under consideration. Dimensional analysis therefore constrains interpretation; it does not prohibit every fractional power.

3.2.2 Why the comparison with one must be dimensionless

Let L be a length and $L_0 > 0$ a reference length. The meaningful numerical variable is

$$x = \frac{L}{L_0}.$$

Changing L_0 changes the numerical value of x. Consequently the statement that $x > 1$ or $x < 1$ depends on the reference scale, as it should: it says that L is larger or smaller than L_0.

For any fixed choice of L_0, however,

$$\left(\frac{L}{L_0}\right)^{1/n} \to 1.$$

Thus the invariant mathematical statement is the convergence of a dimensionless ratio to one. The direction of approach relative to one expresses the chosen normalization.

Remark 3.5. Nondimensionalization is therefore the right language for the original physical observation. The issue is not that a square root of a length is forbidden, but that the numerical comparison of a dimensional quantity with the number one is meaningless until a reference scale has been chosen.

3.3 Complex Principal Roots

3.3.1 Branch convention

For $z \neq 0$, write

$$z = |z|e^{\mathrm{i}\theta}, \qquad \theta = \operatorname{Arg} z \in (-\pi, \pi].$$

The principal logarithm is

$$\operatorname{Log} z = \ln|z| + \mathrm{i}\operatorname{Arg} z,$$

and the principal nth root is

$$z_{\mathrm{pr}}^{1/n} = \exp\left(\frac{1}{n}\operatorname{Log} z\right) = |z|^{1/n} e^{\mathrm{i}\theta/n}.$$

The principal branch is analytic on the slit plane and has the usual branch-cut discontinuity described in standard references such as the NIST Digital Library of Mathematical Functions.

Theorem 3.6 (Principal-root convergence)**.** *For every fixed* $z \in \mathbb{C} \setminus \{0\}$,

$$\lim_{n\to\infty} z_{\mathrm{pr}}^{1/n} = 1.$$

Proof. The real part of $(1/n)\operatorname{Log} z$ tends to zero and its imaginary part also tends to zero. Therefore the exponent tends to zero and the exponential tends to one. ☐

Proposition 3.7 (Uniform version)**.** *Let K be a compact subset of the principal-logarithm domain that does not contain zero. Then*

$$\sup_{z\in K} \left| z_{\mathrm{pr}}^{1/n} - 1 \right| = O\left(\frac{1}{n}\right).$$

Proof. The principal logarithm is bounded on K. Let $M = \sup_{z\in K} |\operatorname{Log} z|$. Then

$$z_{\mathrm{pr}}^{1/n} - 1 = e^{\operatorname{Log} z/n} - 1.$$

On the bounded set of exponents $\operatorname{Log} z/n$, the exponential has a uniform first-order Lipschitz bound for sufficiently large n, giving a constant multiple of M/n. ☐

Proposition 3.8 (First-order complex asymptotics)**.** *For fixed* $z \neq 0$,

$$z_{\mathrm{pr}}^{1/n} = 1 + \frac{\operatorname{Log} z}{n} + O\left(\frac{1}{n^2}\right).$$

Hence

$$n\left(z_{\mathrm{pr}}^{1/n} - 1\right) \to \operatorname{Log} z.$$

3.3.2 The logarithmic-spiral geometry

When $\theta \neq 0$, the polar coordinates of the principal root satisfy

$$r_n = |z|^{1/n}, \qquad \theta_n = \frac{\theta}{n}.$$

If also $|z| \neq 1$, eliminating n gives

$$\ln r_n = \frac{\ln|z|}{\theta}\,\theta_n.$$

Thus the discrete points lie exactly on a logarithmic spiral in polar coordinates. There are two degenerate cases: if $|z| = 1$ and $\theta \neq 0$, the points lie on the unit circle; if $\theta = 0$, they lie on the positive real ray. Hence "spiral toward one" is an exact description in the nondegenerate case, with the circle and ray as limiting special cases.

3.3.3 The full set of roots

Let $R_n(z)$ denote the set of all nth roots of $z \neq 0$. Then

$$R_n(z) = \left\{ |z|^{1/n} \exp\left(\frac{\mathrm{i}(\theta + 2\pi k)}{n}\right) : k = 0, \ldots, n-1 \right\}.$$

Each root has modulus $|z|^{1/n} \to 1$, while the angular spacing is $2\pi/n$.

Theorem 3.9 (Hausdorff convergence of the multivalued roots)**.** *Let $S^1 = \{w \in \mathbb{C} : |w| = 1\}$. For every fixed $z \neq 0$,*

$$d_H\big(R_n(z), S^1\big) \to 0.$$

More precisely,

$$d_H\big(R_n(z), S^1\big) \leq \big||z|^{1/n} - 1\big| + \frac{\pi}{n}.$$

Proof. Every point of $R_n(z)$ has radial distance $||z|^{1/n} - 1|$ from the unit circle, which gives one half of the Hausdorff estimate. Conversely, for any point of S^1, choose the nearest root angle among the equally spaced angles. The angular mismatch is at most π/n, and the chord distance on the unit circle is at most π/n. Adding the radial discrepancy yields the stated bound. □

Remark 3.10. The principal-root sequence selects one point from this increasingly fine discrete polygon. The multivalued problem therefore has a completely different limiting geometry: the selected principal branch converges to one point, whereas the full root set becomes dense around the entire unit circle.

3.4 A Principal-Branch Iterated Attractor

3.4.1 The iteration

Consider

$$z_{k+1} = F(z_k), \qquad F(z) = \frac{\mathrm{i}}{\sqrt{z}},$$

where the square root is principal, and assume $z_0 \neq 0$.

Write

$$z_k = r_k e^{\mathrm{i}\theta_k}, \qquad r_k > 0, \quad \theta_k \in (-\pi, \pi].$$

Then

$$r_{k+1} = r_k^{-1/2}$$

and

$$\theta_{k+1} = \frac{\pi}{2} - \frac{\theta_k}{2}.$$

The second expression automatically lies in $[0, \pi]$, so no further modulo correction is needed after one step.

Theorem 3.11 (Exact orbit formula)**.** *For every $z_0 \neq 0$,*

$$r_k = r_0^{(-1/2)^k}$$

and

$$\theta_k = \frac{\pi}{3} + \left(-\frac{1}{2}\right)^k \left(\theta_0 - \frac{\pi}{3}\right).$$

Consequently,

$$z_k \to e^{\mathrm{i}\pi/3}.$$

Proof. Taking logarithms of the radial recurrence gives

$$\ln r_{k+1} = -\frac{1}{2} \ln r_k,$$

which iterates to the stated formula and implies $r_k \to 1$. For the angular recurrence, subtract $\pi/3$ from both sides:

$$\theta_{k+1} - \frac{\pi}{3} = -\frac{1}{2}\left(\theta_k - \frac{\pi}{3}\right).$$

Iteration gives the exact expression. Since both the radius and angle converge, the complex sequence converges to $e^{\mathrm{i}\pi/3}$. □

Corollary 3.12 (Unique fixed point). *The principal map $F(z) = \mathrm{i}/\sqrt{z}$ has exactly one fixed point in $\mathbb{C} \setminus \{0\}$, namely*

$$z_* = e^{\mathrm{i}\pi/3} = \frac{1}{2} + \frac{\sqrt{3}}{2}\mathrm{i}.$$

Proof. The global convergence theorem implies any fixed point must equal the common limit. Direct substitution verifies that $F(z_*) = z_*$. □

3.4.2 The corrected branch statement

A different starting point does not create a second attractor while the same principal branch is retained. In particular, starting from $z_0 = -\mathrm{i}$ still gives convergence to $e^{\mathrm{i}\pi/3}$.

A conjugate fixed point can be obtained from a different map, for example

$$\widetilde{F}(z) = \frac{-\mathrm{i}}{\sqrt{z}},$$

again with the principal square root. Its angular recurrence has fixed point $-\pi/3$, and therefore its nonzero orbits converge to $e^{-\mathrm{i}\pi/3}$. This is a different dynamical system, not a consequence of the initial condition alone.

Proposition 3.13 (Local multiplier). *On a neighborhood of the fixed point that avoids the branch cut,*

$$F'(z) = -\frac{F(z)}{2z},$$

so at z_*,

$$F'(z_*) = -\frac{1}{2}.$$

Thus the attracting fixed point is locally linearly stable with contraction factor $1/2$.

3.5 Root-Space Synthesis

3.5.1 Relation to the recursive-coordinate programme

The preceding results concern the ordinary root operation and the branch-sensitive map built from the principal square root. The next chapter asks a different structural question: what happens if the successive radicals themselves are promoted to formal coordinates? The distinction is essential. The convergence results here concern points of $\mathbb{C}$, whereas the recursive-coordinate programme introduces a formal ambient vector space together with a realization map back into $\mathbb{C}$.

The elementary function of taking an increasingly high root contains several distinct limiting stories. On positive real inputs it approaches one monotonically with an exact logarithmic first-order term. On negative real inputs, the odd real roots approach minus one, while the principal complex roots approach one. For a fixed nonzero complex input, the principal branch produces a logarithmic-spiral sequence converging to one, whereas the complete set of roots approaches the unit circle in Hausdorff distance. Finally, the nonlinear principal-branch iteration based on the inverse square root has a unique global attractor at the point with argument one third of a full half-turn.

These statements are compatible because they describe different maps, different branch choices, and different objects. The central methodological lesson is therefore not that all roots have one universal limiting behavior, but that the limiting behavior becomes simple once the domain and branch convention are stated before the calculation begins.

Chapter 4

Recursive Imaginary Topology: Corrected Foundation

4.1 The source construction

The source paper defines

$$i_0 = 1, \qquad i_1 = \mathrm{i}, \qquad i_{n+1} = \sqrt{-i_n}, \tag{4.1}$$

and then writes formal combinations

$$Z = \sum_{k=0}^{N} a_k i_k, \qquad a_k \in \mathbb{R}. \tag{4.2}$$

It identifies this collection with an $(N+1)$-dimensional real vector space and proves completeness and path-connectedness. Those two properties are correct only in the formal-coordinate interpretation. The source paper also attempts to define multiplication by taking products in $\mathbb{C}$, while acknowledging that finite closure may fail.

4.2 Formal coordinate space versus complex realization

Definition 4.1 (Formal hyper-imaginary coordinate space)**.** For fixed $N \geq 0$, define

$$\mathcal{H}_N := \mathbb{R}^{N+1}$$

with canonical basis $e_0, \ldots, e_N$. A formal Pulickal coordinate is $v = \sum a_k e_k$.

Proposition 4.2. *$(\mathcal{H}_N, \|\cdot\|_2)$ is complete, path-connected, locally compact, and contractible.*

Proof. These are standard properties of finite-dimensional Euclidean space. Completeness follows from completeness of $\mathbb{R}$ coordinatewise; path-connectedness follows from affine line segments; contractibility is given by $H(v,t) = (1-t)v$. □

Definition 4.3 (Complex realization)**.** Let $\rho_N : \mathcal{H}_N \to \mathbb{C}$ be the real-linear map

$$\rho_N\left(\sum_{k=0}^{N} a_k e_k\right) = \sum_{k=0}^{N} a_k i_k.$$

Theorem 4.4 (Collapse of the literal complex dimension)**.** *For every N, the rank of ρ_N over $\mathbb{R}$ is at most 2. Hence*

$$\dim_{\mathbb{R}}(\mathcal{H}_N / \ker \rho_N) \leq 2.$$

In particular, the recursively generated complex numbers cannot form an $(N+1)$-dimensional linearly independent basis inside $\mathbb{C}$ for $N \geq 2$.

Proof. The codomain $\mathbb{C}$ is a two-dimensional real vector space, so the rank is at most 2. For $N \geq 1$, the images of e_0 and e_1 are 1 and i, which are linearly independent over $\mathbb{R}$ and span $\mathbb{C}$. Hence the rank is exactly 2 and rank-nullity gives $\dim_{\mathbb{R}} \ker \rho_N = N - 1$. The first isomorphism theorem then yields $\mathcal{H}_N / \ker \rho_N \cong \operatorname{im} \rho_N = \mathbb{C}$ as real vector spaces. For $N = 0$, the image is $\mathbb{R}$. ☐

Corollary 4.5 (Exact quotient dimension). *For $N = 0$, $\mathcal{H}_0 / \ker \rho_0 \cong \mathbb{R}$. For every $N \geq 1$, $\dim_{\mathbb{R}} \ker \rho_N = N - 1$ and $\mathcal{H}_N / \ker \rho_N \cong \mathbb{C}$ as real vector spaces. Thus the entire finite hierarchy of literal recursive coordinates collapses canonically to the same two-dimensional real quotient as soon as the coordinates 1 and i are present.*

Proof. This is the rank-nullity conclusion in the preceding theorem together with the first isomorphism theorem. ☐

Salvage

The mathematically clean interpretation is therefore a *formal coordinate space with a complex realization map.* The kernel records redundancy among recursively generated coordinates. This is a legitimate linear-algebraic object even though it is not a new number system.

4.3 Why the algebra claim needs repair

Suppose one tries to define

$$e_j e_k = \sum_{m=0}^{N} C_{jk}^{m} e_m$$

by requiring the realization to agree with multiplication in $\mathbb{C}$. This can only be done intrinsically if the proposed multiplication closes in $\mathcal{H}_N$ and is compatible with $\ker \rho_N$.

Proposition 4.6 (Descent criterion). *Let V be a vector space with a bilinear multiplication $m : V \times V \to V$, and let $\rho : V \to W$ be linear. A multiplication on $\operatorname{im} \rho$ induced by ρ is well-defined by*

$$\rho(v) \star \rho(w) := \rho(vw)$$

only if $\ker \rho$ is a two-sided ideal of V.

Proof. If $v - v' \in \ker \rho$, well-definedness requires $\rho((v - v')w) = 0$ for every w, and similarly on the right. These are exactly the ideal conditions. ☐

Theorem 4.7 (Canonical quotient consequence). *For $N \geq 1$, the complex-realization quotient*

$$\mathcal{H}_N / \ker \rho_N$$

is isomorphic as a real vector space to $\operatorname{im} \rho_N$, hence has dimension at most 2. If a bilinear multiplication on $\mathcal{H}_N$ makes ρ_N an algebra homomorphism into $\mathbb{C}$, then $\ker \rho_N$ is a two-sided ideal and the induced quotient algebra is isomorphic to $\mathbb{C}$ for every $N \geq 1$. In particular, no such compatible quotient can produce a genuinely new $(N + 1)$-dimensional real number system inside $\mathbb{C}$.

Proof. The vector-space statement follows from the first isomorphism theorem. If ρ_N is an algebra homomorphism, its kernel is a two-sided ideal, and the universal property of the quotient gives an injective algebra homomorphism

$$\mathcal{H}_N / \ker \rho_N \longrightarrow \mathbb{C}$$

with image $\operatorname{im} \rho_N = \mathbb{C}$ for $N \geq 1$. ☐

The original paper does not establish this compatibility, and the finite set $\{i_0, \ldots, i_N\}$ is not automatically closed under complex multiplication. Thus commutativity and associativity of $\mathbb{C}$ cannot simply be transferred to an unconstructed finite algebra. The strongest canonical object obtained from the literal realization is instead the quotient above.

4.4 Correct principal-branch dynamics

Write

$$i_n = r_n e^{\mathrm{i}\theta_n}, \qquad r_n > 0, \quad \theta_n \in (-\pi, \pi].$$

For the principal square root,

$$r_{n+1} = \sqrt{r_n}.$$

Hence

$$r_n = r_0^{2^{-n}} \longrightarrow 1. \tag{4.3}$$

In particular, the recursively generated basis orbit itself starts from $|i_0| = 1$ and therefore satisfies $|i_n| = 1$ for every n. For general nonzero initial data, the radial component is attracted to one.

The angular dynamics require the principal argument of $-z$. Thus

$$\theta_{n+1} = \begin{cases} \frac{\theta_n + \pi}{2}, & \theta_n \le 0, \\ \frac{\theta_n - \pi}{2}, & \theta_n > 0. \end{cases} \tag{4.4}$$

Theorem 4.8 (Two-cycle for the principal recursive-root orbit). *For every initial $z_0 \neq 0$ whose argument is in $(-\pi, \pi]$, the principal iteration*

$$z_{n+1} = \sqrt{-z_n}$$

has radial component $|z_n| \to 1$. If $\theta_0 > 0$, the even and odd angular subsequences converge respectively to

$$\frac{\pi}{3}, \qquad -\frac{\pi}{3},$$

and if $\theta_0 < 0$ the parity is reversed. For $\theta_0 = 0$ or π, the first one or two iterates enter one of these two regimes, after which the same conclusion holds. Thus the nonzero principal orbit approaches the two-cycle

$$e^{\mathrm{i}\pi/3} \longleftrightarrow e^{-\mathrm{i}\pi/3}.$$

Proof. If $\theta > 0$, one application of (4.4) gives a negative angle and two applications give

$$T^2(\theta) = \frac{\theta + \pi}{4}.$$

This affine map is a contraction with ratio $1/4$ and fixed point $\theta = \pi/3$. For negative θ, the corresponding two-step map is

$$T^2(\theta) = \frac{\theta - \pi}{4},$$

with fixed point $-\pi/3$. Contraction gives convergence of the appropriate parity subsequences. The radial conclusion follows from $r_n = r_0^{2^{-n}}$. □

Mathematical boundary

The source claim that -1 is an attracting fixed point of the principal map is false: $\sqrt{-(-1)} = \sqrt{1} = 1$. Consequently the source identification of the Julia set as the unit circle is not a valid theorem for the principal square-root branch. The corrected object is the branch-sensitive piecewise-affine angular dynamical system above.

Proposition 4.9 (The principal branch is not globally continuous). *Let $f(z) = \sqrt{-z}$ with the principal square root. Then f is continuous on its natural slit domain but is not continuous across the positive real axis. In particular, as $z \to 1$ from the two half-planes, the limiting values of $f(z)$ are opposite imaginary numbers. Therefore the principal recursive-root dynamics are not a globally continuous self-map of $\mathbb{C}$ with the Euclidean topology.*

Proof. The principal square root is defined by $\sqrt{re^{i\theta}} = \sqrt{r}\, e^{i\theta/2}$ for $\theta \in (-\pi, \pi]$. When z approaches 1 from opposite sides, $-z$ approaches the negative real axis with arguments approaching $-\pi$ and π, respectively. The corresponding square roots approach $-i$ and $+i$. Hence the two one-sided limits disagree. □

Remark 4.10 (On the word "Julia")**.** Classical Julia sets are most canonically defined for globally specified holomorphic dynamical systems such as rational maps on the Riemann sphere. The principal map above is a branch-defined map with a slit and a jump across a branch-cut preimage. Its angular two-cycle is a genuine dynamical result, but calling its unit circle a "Julia set" requires an additional definition of Julia-type behavior for this branch-sensitive system. The source claim therefore cannot be recovered merely by renaming the corrected orbit dynamics.

4.5 Relation to the Approaching Unity chapter

The preceding "Approaching Unity" chapter studies ordinary root operations as functions or finite sets in $\mathbb{C}$: high-order roots converge to unity on the principal branch, the full root sets approach the unit circle in Hausdorff distance, and the principal map $z \mapsto \mathrm{i}/\sqrt{z}$ has a unique attracting fixed point. The present chapter asks a different question. It promotes the successive recursively generated radicals to formal coordinates and then studies the realization map into $\mathbb{C}$. The two programmes therefore complement rather than duplicate one another.

4.6 Notebook-I relation

This material overlaps with the mature recursive-root analysis already placed in Notebook I. In Notebook III it is retained as corrected structural context rather than as a competing foundational theory. The genuinely new contribution here is the explicit realization-map viewpoint and the descent/closure criterion for any proposed finite algebra.

Chapter 5

From Ellipse Heuristics to Exact Ellipsoid Certificates

5.1 The historical 2D programme

The earlier PBP-0024 work began with a practical rule: compare distances from a candidate point to selected extreme points of an ellipse against corresponding axis lengths. The paper itself repeatedly uses heuristic language for this construction and later adds rotated and simplified algorithms. The heuristic is worth preserving as the origin of a more exact geometric certificate.

5.2 The correct geometric problem

Let

$$E = \left\{ x \in \mathbb{R}^d : \sum_{i=1}^{d} \frac{x_i^2}{a_i^2} \le 1 \right\}, \qquad a_i > 0.$$

For a principal vertex $v_j^- = -a_j e_j$, define its exact farthest-point radius

$$M_j := \max_{x \in E} \left\| x - v_j^- \right\|_2 .$$

Then

$$\left\| p - v_j^- \right\| > M_j \quad \Longrightarrow \quad p \notin E.$$

The interesting question is therefore to calculate M_j exactly rather than assume that it always equals $2a_j$.

Theorem 5.1 (Exact farthest distance from a principal vertex)**.** *Let*

$$A := \max_{1 \le i \le d} a_i^2, \qquad B := a_j^2.$$

Then

$$M_j^2 = \begin{cases} 4B, & A \le 2B, \\ \dfrac{A^2}{A - B}, & A > 2B. \end{cases} \tag{5.1}$$

Proof. Write $x_i = a_i y_i$ with $\sum y_i^2 \le 1$. For $v_j^- = -a_j e_j$,

$$\left\| x - v_j^- \right\|^2 = B + \sum_i a_i^2 y_i^2 + 2B y_j.$$

For $t = y_j$, using $a_i^2 \le A$ for $i \ne j$ gives

$$\sum_i a_i^2 y_i^2 \le Bt^2 + A(1 - t^2).$$

Therefore

$$\left\|x - v_j^-\right\|^2 \le A + B + (B - A)t^2 + 2Bt =: q(t), \quad -1 \le t \le 1.$$

If $A = B$, $q(t) = 2B + 2Bt$ and its maximum is $4B$ at $t = 1$. If $A > B$, q is concave and its stationary point is

$$t_* = \frac{B}{A - B}.$$

When $A \le 2B$, $t_* \ge 1$ and the maximum on $[-1, 1]$ is again at $t = 1$, giving $4B$. When $A > 2B$, $t_* \in (0, 1)$ and

$$q(t_*) = A + B + \frac{B^2}{A - B} = \frac{A^2}{A - B}.$$

The bounds are attained by placing all remaining mass in a coordinate i_* with $a_{i_*}^2 = A$, and therefore the formula is exact. □

Corollary 5.2 (When the original $2a_j$ rule is exact)**.** *The simple threshold $M_j = 2a_j$ is exact precisely when the longest squared semi-axis satisfies*

$$A \le 2a_j^2.$$

In highly anisotropic ellipsoids it can fail for short axes.

Example 5.3. For $(a, b, c) = (3, 2, 1)$ and the b-axis vertex, $A = 9$, $B = 4$, so

$$M_b^2 = \frac{81}{5} = 16.2.$$

Thus the correct exclusion threshold is $\sqrt{16.2}$, not $2b = 4$.

5.3 The 2D predecessor is recovered

For an ellipse with semi-axes $a \ge b$, the theorem yields

$$M_a = 2a$$

for a major-axis vertex, while for a minor-axis vertex

$$M_b^2 = \begin{cases} 4b^2, & a^2 \le 2b^2, \\ \dfrac{a^4}{a^2 - b^2}, & a^2 > 2b^2. \end{cases}$$

Thus the 2D and 3D programmes become one theorem rather than two unrelated heuristics.

5.4 Rotated ellipsoids

Let

$$E_{\mu,\Sigma} = \{x : (x - \mu)^T \Sigma^{-1} (x - \mu) \le 1\},$$

where Σ is symmetric positive definite. Write

$$\Sigma = Q \operatorname{diag}(a_1^2, \ldots, a_d^2) Q^T.$$

The map $y = Q^T(x - \mu)$ preserves Euclidean distances because Q is orthogonal and removes only the translation by μ. Hence the farthest-distance theorem applies in the principal coordinates and can be transported back exactly; the principal vertices are $v_j^{\pm} = \mu \pm a_j q_j$, where q_j is the j-th column of Q.

Salvage

The corrected theorem is stronger than the original claim: it produces a mathematically certified sufficient condition for exteriority without pretending to give a complete anomaly classifier. The statistical problem of estimating μ, A, and the threshold from noisy data is a separate empirical problem.

5.5 Certificate architecture

Given anchor points $V = \{v_1, \dots, v_m\}$ and exact radii M_j, define

$$C(p) = \max_j \big(\|p - v_j\| - M_j\big).$$

Then

$$C(p) > 0 \Longrightarrow p \notin E.$$

The condition $C(p) > 0$ is a one-sided certificate. For the selected anchors, the exact unresolved region is

$$\left(\bigcap_j \overline{B}(v_j, M_j)\right) \setminus E,$$

which need not be empty. Thus the certificate is sound for exteriority but is not, without additional hypotheses, a complete membership classifier.

Research programme 5.4 (General certified exclusion)**.** For compact convex sets $K \subset \mathbb{R}^d$, determine small anchor families and computable bounds $M_j \geq \max_{x \in K} \|x - v_j\|$ that minimize evaluation cost subject to a prescribed unresolved-volume or false-negative tolerance. The ellipsoid theorem provides an exact benchmark case.

Chapter 6

Homotopy Type Theory and Geometric Equality: Shape Quotients for Euclidean Triangles

6.1 Placement within Volume III

This chapter supplies the quotient-theoretic counterpart to the geometric-certificate programme. The ellipsoid chapters use distances and inequalities to certify membership or exteriority; the present chapter addresses a different geometric question: when should two represented triangles be regarded as the same shape? The answer is obtained by separating strict equality from congruence and similarity, then representing those equivalence relations by set quotients. This makes the chapter a direct bridge from geometric data to the finite, invariant representations developed later in the volume.

6.1.1 Several notions of sameness

A located triangle has more structure than its intrinsic shape. Its location, vertex ordering, orientation, and scale can all be recorded in the representation. Ordinary mathematical language then uses several different notions of sameness. Two located triangles may be literally the same object, congruent but translated or rotated, or merely similar after a change of scale.

The point of a type-theoretic treatment is not to collapse these notions into one relation. It is to place them at different levels and then state precisely when an equivalence relation is represented by equality in a quotient object.

6.1.2 Four levels of interpretation

It is useful to separate four objects:

1. the type of located labeled triangles;
2. the congruence relation on that type;
3. the quotient of triangles by congruence;
4. the quotient of triangles by similarity.

The third object retains absolute scale but forgets location and labeling to the extent encoded by the chosen congruence relation. The fourth also forgets scale.

Remark 6.1. The word "moduli" must be used carefully. A set of equivalence classes, a topological quotient, a homotopy type, and a quotient stack are related but different objects. This paper constructs the first rigorously and explains how the others arise from additional structure.

6.2 Type-theoretic foundations

6.2.1 Types, identity, and sets

In Homotopy Type Theory, for a type A and terms $a, b : A$, the identity type $a = b$ is itself a type. A type is an h-set when all its identity types are propositions. Euclidean space, viewed as a type, is an h-set in standard HoTT foundations.

A higher path can exist between equality proofs in a general type, so a higher inductive definition that adds paths does not automatically produce a set. This observation is central to the quotient construction below.

6.2.2 Sigma types and propositional fibers

Suppose A is a set and $P : A \to \mathrm{Prop}$. Then the dependent sum

$$\sum_{a:A} P(a)$$

is again a set. The proposition-valued fiber records data that is proof-irrelevant at the level of the resulting object.

This will allow a triangle to contain its nondegeneracy proof without introducing spurious identity distinctions.

6.2.3 Univalence

For types A, B in a univalent universe $\mathcal{U}$, univalence gives an equivalence

$$(A =_{\mathcal{U}} B) \simeq (A \simeq B).$$

Its role is to identify equivalent types as equal types in the universe. It does not, by itself, say that an arbitrary path constructor added to a type is the only source of equality in that type.

6.3 Located Euclidean triangles

6.3.1 The point type

Take

$$\mathsf{Point} := \mathbb{R}^2.$$

Since $\mathbb{R}$ is a set and finite products of sets are sets, Point is a set. In particular, for points A, B, the identity type $A = B$ expresses literal equality of points, not the existence of a geometric segment between them.

Remark 6.2 (On geometric segments)**.** The identity type of two points should not be called the geometric segment joining those points. For distinct points it is empty. A geometric segment is a different object, for example the image of the interval under affine interpolation. The identity type is nevertheless exactly the correct object for strict equality of vertices.

6.3.2 Non-collinearity

For points $A, B, C : \mathbb{R}^2$, define collinearity by the existence of a scalar t satisfying

$$C - A = t(B - A).$$

Non-collinearity is the negation of this proposition.

Definition 6.3 (Triangle)**.** A labeled located triangle is the dependent sum

$$\text{Triangle} := \sum_{(v_1,v_2,v_3):\mathsf{Point}^3} \mathsf{NonCollinear}(v_1, v_2, v_3).$$

Proposition 6.4. Triangle *is a set.*

Proof. The base Point^3 is a set and non-collinearity is proposition-valued. A sigma type over a set with proposition-valued fibers is a set. □

6.3.3 Strict equality

For triangles T, U : Triangle, strict equality is simply

$$T = U.$$

Because the vertex space is a set and the non-collinearity proof is proposition-valued, equality of triangles reduces to equality of their vertex triples. Thus a translation, rotation, or relabeling is not strict equality of located triangles.

6.4 Congruence as an equivalence relation

6.4.1 Side-length data

For a triangle $T = (v_1, v_2, v_3, p)$ define the ordered side-length triple

$$\ell(T) = \big(\operatorname{dist}(v_1, v_2), \operatorname{dist}(v_2, v_3), \operatorname{dist}(v_3, v_1)\big).$$

Let S_3 act by relabeling the three vertices; this induces the corresponding permutation action on the three side lengths. Every permutation of the three sides arises from a permutation of the vertices. The use of all permutations means that congruence here is the ordinary unlabeled Euclidean notion, so reflections are permitted.

Definition 6.5 (Congruence)**.** Define

$$\operatorname{Cong}(T, U) := \left\| \sum_{\sigma:S_3} \ell(T) = \sigma \cdot \ell(U) \right\|,$$

where the outer bars denote propositional truncation. Informally, T and U are congruent when their side-length triples agree after a permutation.

The propositional truncation is important. There may be several permutations witnessing congruence, especially for symmetric triangles, but congruence itself is a proposition: either the two triangles are congruent or they are not.

Proposition 6.6 (Congruence is reflexive)**.** *For every T,* $\operatorname{Cong}(T, T)$.

Proof. Use the identity permutation and reflexivity of equality of real numbers. □

Proposition 6.7 (Congruence is symmetric)**.** *If* $\operatorname{Cong}(T, U)$ *then* $\operatorname{Cong}(U, T)$.

Proof. Any witnessing permutation σ can be replaced by its inverse. Equality is symmetric, so the corresponding side-length equalities reverse. □

Proposition 6.8 (Congruence is transitive)**.** *If* $\operatorname{Cong}(T, U)$ *and* $\operatorname{Cong}(U, V)$ *then* $\operatorname{Cong}(T, V)$.

Proof. Compose the two witnessing permutations and compose the corresponding equality proofs. □

Thus Cong is a proposition-valued equivalence relation on Triangle.

6.4.2 Relation with classical SSS congruence

The side-length formulation is not an arbitrary coding trick. In the Euclidean plane, two nondegenerate triangles with the same three side lengths are related by a Euclidean isometry, after a suitable correspondence of their vertices. This is the classical SSS theorem.

Theorem 6.9 (Euclidean congruence criterion)**.** *For nondegenerate labeled triangles T, U, the following are equivalent:*

(i) $\mathrm{Cong}(T, U)$*;*

(ii) after a permutation of vertices, there exists a Euclidean isometry carrying the vertices of T to those of U.

Proof sketch. After permuting vertices so corresponding sides have equal lengths, place the first triangle with two vertices at $(0, 0)$ and $(c, 0)$. The third vertex is determined, up to reflection across the first axis, by the two remaining side lengths through the law of cosines. The same construction for the second triangle therefore yields the same coordinates up to an orthogonal transformation and translation. These transformations are Euclidean isometries. The converse is immediate because isometries preserve distances. □

6.5 The correct quotient object

6.5.1 Why a naive path HIT is insufficient

A tempting definition is to introduce a point constructor

$$q : \mathrm{Triangle} \to Q$$

and, for each congruence proof, add a path

$$q(T) = q(U).$$

This does establish that congruent triangles become equal in Q. It does not, by itself, prove the converse statement that every equality in Q arises from congruence.

The reason is structural. A general higher inductive type can have nontrivial higher identity types, and its path spaces can contain composites, inverses, and higher coherence not present in the original relation. The HoTT literature explicitly distinguishes general quotients from set-quotients and notes that the latter are truncated to sets. [19, 20]

6.5.2 Set quotient

Let A be a type and $R : A \to A \to \mathrm{Prop}$ an equivalence relation. The set quotient A/R is a type equipped with a projection

$$q : A \to A/R,$$

paths identifying $q(a)$ and $q(b)$ whenever $R(a, b)$ holds, together with the requirement that A/R is a set.

Its key universal property is that maps from A/R into any set B are exactly maps from A to B that respect R. [19]

Definition 6.10 (Congruence shape)**.** Define

$$\mathrm{Shape}_{\mathrm{cong}} := \mathrm{Triangle} \,/\, \mathrm{Cong}\,.$$

The projection is denoted q_{cong}.

Theorem 6.11 (Effectiveness of the congruence quotient)**.** *For all T, U :* Triangle*,*

$$\mathrm{Cong}(T, U) \simeq \big(q_{\mathrm{cong}}(T) = q_{\mathrm{cong}}(U)\big).$$

Proof sketch. The forward map is supplied by the quotient constructor. For the reverse implication, use the standard effectiveness theorem for set quotients: when R is an equivalence relation, the canonical projection $q : A \to A/R$ satisfies $R(a, b) \simeq (q(a) = q(b))$. Applying this result with $A = \mathrm{Triangle}$ and $R = \mathrm{Cong}$ gives the stated equivalence. The role of the set-truncation is precisely to rule out additional higher identity information in the quotient. [19] □

This is the corrected form of the central claim: equality in the quotient really is congruence, but this requires the quotient construction and its set-truncation/universal property rather than univalence alone.

6.5.3 Set-truncated HIT presentation

The same object may be presented schematically by a higher inductive type with constructors

$$\begin{aligned}\mathsf{shape} &: \mathrm{Triangle} \to \mathrm{Shape}_{\mathrm{cong}},\\ \mathsf{congPath} &: \mathrm{Cong}(T, U) \to \big(\mathsf{shape}(T) = \mathsf{shape}(U)\big),\end{aligned}$$

together with an explicit 0-truncation condition asserting that $\mathrm{Shape}_{\mathrm{cong}}$ is a set. This is the appropriate HIT formulation for a set quotient. [20]

6.6 Similarity is a different quotient

6.6.1 Definition

Congruence retains scale. Similarity does not.

Definition 6.12 (Similarity)**.** For triangles T, U, define $\mathrm{Sim}(T, U)$ by propositional truncation of the type containing a permutation $\sigma \in S_3$, a positive real scale factor λ, and proofs

$$\ell(T) = \lambda\, \sigma \cdot \ell(U).$$

Proposition 6.13. Sim *is an equivalence relation.*

Proof. Reflexivity uses $\lambda = 1$ and the identity permutation. Symmetry uses λ^{-1} and the inverse permutation. Transitivity multiplies scale factors and composes permutations. □

Definition 6.14 (Similarity shape)**.** Define

$$\mathrm{Shape}_{\mathrm{sim}} := \mathrm{Triangle} \,/\, \mathrm{Sim}\,.$$

Theorem 6.15 (Similarity is not congruence)**.** *There exist triangles that are similar but not congruent.*

Proof. Scale any nondegenerate triangle by a factor $\lambda \neq 1$. Side ratios remain equal, so the triangles are similar. Their side lengths are not all equal, so they are not congruent. □

This is why a congruence quotient cannot simultaneously be described as a quotient "ignoring scale." Scale must be removed by a separate similarity quotient or a normalization.

6.7 A concrete triangle-shape moduli space

6.7.1 Normalized side lengths

For a nondegenerate triangle let

$$(a, b, c) \in \mathbb{R}^3_{>0}$$

be its side lengths and let $p = a + b + c$. Define normalized lengths

$$\widehat{a} = \frac{a}{p}, \qquad \widehat{b} = \frac{b}{p}, \qquad \widehat{c} = \frac{c}{p}.$$

Then

$$\widehat{a} + \widehat{b} + \widehat{c} = 1$$

and the strict triangle inequalities remain valid. Conversely, any positive triple summing to one and satisfying the strict triangle inequalities determines a similarity class.

Define

$$\Delta_{\triangle} := \left\{(a, b, c) \in \mathbb{R}^3_{>0} : a + b + c = 1,\ a < b + c,\ b < c + a,\ c < a + b\right\}.$$

The symmetric group S_3 acts by permuting coordinates.

Theorem 6.16 (Concrete similarity-shape realization)**.** *The set of unlabeled similarity classes of nondegenerate Euclidean triangles is naturally identified with the orbit space*

$$\Delta_{\triangle}/S_3.$$

Proof. Normalization removes the positive scale factor. Two triangles have the same normalized side-length multiset exactly when their side-length triples differ by a permutation and a common positive factor. By SSS, this is equivalent to similarity. □

This is a concrete set-theoretic moduli model of similarity classes. It is two-dimensional because the equation $a + b + c = 1$ removes one degree of freedom from three side lengths; the strict triangle inequalities restrict the resulting open region, and quotienting by the finite permutation group removes labeling. If one additionally equips the quotient with the topology induced from the normalized region, one obtains a topological moduli space; this paper does not require the stronger stack-theoretic structure.

The literature contains several other realizations of the same similarity moduli idea. For example, an open disk can serve as a moduli space of similarity classes of triangles, and the three-body problem literature distinguishes shape space of congruence classes from the shape sphere of oriented similarity classes. [22, 23]

6.7.2 Congruence shape space

Without normalization, ordered positive side lengths satisfying the strict triangle inequalities form an open cone. Quotienting by S_3 gives a concrete side-length model of congruence classes. Similarity is obtained from that cone by quotienting its positive radial scaling action.

This yields a useful hierarchy:

$$\text{located triangles} \longrightarrow \text{congruence classes} \longrightarrow \text{similarity classes.}$$

6.8 What HoTT contributes

6.8.1 Equality after quotienting

The important conceptual move is now precise. Strict equality lives in Triangle. Congruence is an equivalence relation on Triangle. The quotient turns that relation into identity of quotient classes.

The resulting statement is not "congruent objects are literally the same object." They are different elements of Triangle whose images become equal in a different type. This is the mathematically correct sense in which an equivalence can be represented by equality.

6.8.2 Why univalence is still relevant

Univalence becomes important when the objects under study themselves carry type-level structure. Equivalent types may be transported as equal types, and equivalences between structured objects can be promoted to equalities in suitable universes. But the geometric quotient theorem above does not depend on pretending that univalence constructs the quotient.

The distinction is useful:

1. quotient constructors specify which geometric objects are identified;

2. set truncation controls the higher identity structure of the quotient;
3. univalence governs equality of types and supports transport along type equivalences.

6.8.3 Orientation and automorphisms

The present quotient treats reflections as congruences. If orientation is part of the intended structure, the equivalence relation must be changed to use orientation-preserving isometries or even oriented vertex correspondences.

Likewise, the set quotient deliberately forgets stabilizer information. An equilateral triangle has nontrivial symmetries, and a groupoid or stack quotient can retain those automorphisms. Thus a set quotient is a shape classifier, not a full moduli stack.

Remark 6.17. This distinction is not merely terminological. In contemporary moduli problems, preserving automorphisms can be essential. A finite orbit set, a topological quotient, and a quotient stack all encode different information about the same underlying group action.

6.9 Formal consequences

6.9.1 Properties that descend through the quotient

Let $P : \mathrm{Triangle} \to \mathrm{Prop}$ be a property invariant under congruence:

$$\mathrm{Cong}(T, U) \to (P(T) \leftrightarrow P(U)).$$

Then P descends to a well-defined proposition on $\mathrm{Shape}_{\mathrm{cong}}$. In informal terms, any property depending only on intrinsic Euclidean shape can be stated at the quotient level.

Proposition 6.18 (Descent of congruence-invariant properties)**.** *If P is proposition-valued and congruence-invariant, there exists a unique predicate $\overline{P} : \mathrm{Shape}_{\mathrm{cong}} \to \mathrm{Prop}$ such that*

$$\overline{P}(q_{\mathrm{cong}}(T)) \leftrightarrow P(T).$$

Proof. Use the set-quotient elimination principle into the proposition-valued codomain Prop, verifying that P respects the quotient relation. [19] □

This is one of the practical benefits of the quotient formulation. The theorem is not merely philosophical: it tells us exactly which geometric statements can be moved from located objects to shape classes.

6.9.2 Similarity-invariant properties

The same argument applies to $\mathrm{Shape}_{\mathrm{sim}}$. A property such as “has two equal angles” descends to similarity classes, while a property such as “has perimeter three” does not.

6.10 Formalization blueprint

6.10.1 Proof-assistant representation

A future formalization in Cubical Agda, Lean, or another HoTT-capable environment can follow the order:

1. define points and equality;
2. define non-collinearity as a proposition;
3. define the dependent triangle type;
4. define side lengths;

5. define the proposition-valued congruence relation and prove equivalence;
6. construct the set quotient;
7. prove effectiveness of quotient equality;
8. define similarity and its quotient;
9. construct the normalized side-length equivalence with the similarity quotient;
10. formalize descent of invariant properties.

The formal development should treat geometric theorems such as SSS as explicit lemmas rather than silently identifying side-length equalities with Euclidean isometries.

6.10.2 What would count as a genuine implementation result

A completed proof-assistant development should provide machine-checked definitions and proofs of the core equivalences. This paper does not claim that such a development has already been completed. Its result is instead a mathematically coherent target whose missing formal steps are explicit.

6.11 Applications and limits

6.11.1 Geometry education

The framework gives a precise vocabulary for distinguishing identity of located objects from equivalence of intrinsic shape. This may be pedagogically useful, but the paper does not claim a measured educational improvement.

6.11.2 Automated theorem proving

A theorem that descends through the congruence quotient can be stated at the shape level. In a proof assistant, transport and quotient elimination can make this separation computationally explicit. The benefit is a formalization hypothesis rather than a demonstrated performance result.

6.11.3 CAD and spatial reasoning

CAD systems often distinguish an object instance from its invariant geometric form. A quotient representation is conceptually aligned with this distinction. To turn the idea into an engineering result one would need a concrete data structure, algorithms, and benchmarks.

6.11.4 AI and robotics

An AI system that should recognize shape independently of location or rigid motion can be modeled using invariants or orbit representations. The present HoTT framework provides a specification language for the desired equivalence, not an already-implemented recognition system.

6.12 Verification notes inherited from the standalone preprint

6.12.1 A short proof of the SSS coordinate lemma

Let two nondegenerate triangles have side lengths a, b, c, with c chosen as the distance between the first two vertices. Place those vertices at $(0, 0)$ and $(c, 0)$. If the third vertex is (u, v), then

$$u^2 + v^2 = b^2, \qquad (u - c)^2 + v^2 = a^2.$$

Subtracting gives

$$2cu = b^2 + c^2 - a^2,$$

so u is uniquely determined. Then $v^2 = b^2 - u^2$ is determined, and nondegeneracy gives $v \neq 0$. The two possibilities $v = \pm\sqrt{v^2}$ are exchanged by reflection in the first coordinate axis. Hence the two triangles are related by an isometry.

6.12.2 Quotient versus higher quotient

A general quotient higher inductive type with point and path constructors retains the freely generated path structure unless an additional truncation is imposed. A set quotient adds the requirement that all parallel paths between two points are equal. This is the technical reason that the corrected quotient construction is stronger than the original path-generator definition. [20, 21]

6.12.3 Reference convention for the root chapter

Throughout the root material, the principal logarithm uses $\operatorname{Arg} z \in (-\pi, \pi]$ when boundary values on the negative real axis are required, and the principal power is defined through that logarithm. Multivalued roots are treated as finite sets rather than as a single-valued function.

6.13 Notebook-III Conclusion: Equality, Quotients, and Shape

The central contribution of this reconstruction is a precise separation of three ideas that are often conflated. A located triangle is an element of a type. Congruence is an equivalence relation on that type. A quotient by congruence turns that relation into equality of quotient classes. Similarity requires a different quotient because it removes scale.

The naive statement that a higher-inductive path constructor automatically makes congruence proofs equivalent to quotient identities is too strong. General higher inductive types may contain additional higher homotopy, so the desired effectiveness property must be obtained from an appropriate quotient construction and, when a set of shapes is intended, explicit set truncation. [19, 20]

The resulting picture is coherent. Strict equality belongs to the original object type; congruence belongs to the quotient relation; similarity belongs to a coarser quotient; and the concrete normalized side-length region provides a familiar geometric realization of similarity classes. Univalence remains valuable as the principle connecting equivalence and equality at the level of types, but it should not be assigned work that belongs to quotient elimination and truncation.

The most promising next step is a machine-checked formalization of the core quotient theorem together with explicit Euclidean lemmas. Once those are formalized, the applications to automated geometry and shape reasoning can be evaluated as actual computational systems rather than as conceptual analogies.

Chapter 7

Finite-Resolution Number Spaces

7.1 What survives from "The Number of Numbers"

The source paper correctly senses a distinction between mathematical infinity and finite physical or computational resources, but its quantitative construction equates Euclidean volume with the number of real numbers in a region and introduces arbitrary constants as "practical infinity." Those steps are not valid mathematical derivations.

Mathematical boundary

A finite-volume region of $\mathbb{R}^d$ still contains uncountably many real points. Volume measures geometric size; cardinality measures set size. They must not be identified.

7.2 Operational replacement

A computational system does not represent every real number. It represents a finite collection of codes, quantization cells, floating-point values, or other states. This suggests a precise finite-resolution model.

Definition 7.1 (Resolution-constrained representation)**.** Let (K, d) be a compact metric space and let $Q : K \to \mathcal{Q}$ be a finite-valued representation map. Suppose each code $q \in \mathcal{Q}$ has a chosen representative $\hat{x}_q \in K$. The represented resolution is controlled by

$$\varepsilon_Q := \sup_{x \in K} d(x, \hat{x}_{Q(x)}),$$

and the operational number of distinguishable states is $|\mathcal{Q}|$.

The correct mathematical complexity measure is often a covering number. For $\varepsilon > 0$,

$$N(K, \varepsilon, d)$$

is the minimum number of metric balls of radius ε required to cover K, and the metric entropy is

$$H_\varepsilon(K) = \log N(K, \varepsilon, d).$$

Covering numbers are standard tools for discretizing infinite hypothesis classes and quantifying finite-resolution complexity. [14]

Proposition 7.2 (Box count)**.** *For a d-dimensional box $[-R, R]^d$ and a uniform grid anchored at $-R$ with spacing ε, the number of grid points lying in the box is*

$$N_{\text{grid}} = \left(\left\lfloor \frac{2R}{\varepsilon} \right\rfloor + 1 \right)^d.$$

When $2R/\varepsilon$ is an integer, the grid contains both endpoints in every coordinate direction.

Proposition 7.3 (Elementary covering bounds for a Euclidean ball). *For the closed Euclidean ball $B_R^d \subset \mathbb{R}^d$, there are constants depending only on d such that*

$$c_d \left(\frac{R}{\varepsilon}\right)^d \leq N(B_R^d, \varepsilon) \leq C_d \left(1 + \frac{R}{\varepsilon}\right)^d$$

for $0 < \varepsilon \leq R$.

Proof. The upper bound follows by covering the ball by grid cells of diameter at most a constant multiple of ε. For the lower bound, choose a maximal ε-separated subset; the disjoint balls of radius $\varepsilon/2$ around its points fit into a slightly enlarged ball, giving the standard packing-to-covering comparison. □

Thus the asymptotic scaling is fundamentally

$$N(B_R^d, \varepsilon) = \Theta((R/\varepsilon)^d)$$

up to dimension-dependent constants.

Proposition 7.4 (Finite codebook entropy bound). *If $Q : K \to \mathcal{Q}$ has finite image and each code is stored by a fixed-length binary word of length m, then $|\mathcal{Q}| \leq 2^m$ and*

$$H_0 := \log_2 |\mathcal{Q}| \leq m.$$

Thus the finite codebook size, rather than Euclidean volume, is the correct combinatorial quantity for a fixed binary representation.

Proof. There are at most 2^m binary strings of length m. Taking $\log_2$ gives the result. □

Theorem 7.5 (Codebook lower bound from covering numbers). *Let (K, d) be compact and suppose a representation uses at most M representatives with uniform error at most ε. Then*

$$N(K, \varepsilon, d) \leq M.$$

Consequently, if $N(K, \varepsilon, d) > M$, no M-state representation can achieve uniform error ε.

Proof. The M representatives define an ε-cover of K. By minimality of the covering number, at least $N(K, \varepsilon, d)$ representatives are required. □

7.3 Operational zero

The source intuition that a physical or numerical zero should depend on resolution can be made precise without denying the existence of mathematical zero.

Definition 7.6 (Operational zero). Assume $0 \in K$ and let $Q : K \to \mathcal{Q}$ be a representation map. The operational zero is the code cell

$$Z_Q := Q^{-1}(Q(0)).$$

All states in Z_Q are indistinguishable from zero under that representation.

Definition 7.7 (Operational infinity). For a representation architecture with finite dynamic range, define

$$R_Q := \sup\{\|\hat{x}_q\| : q \in \mathcal{Q}\}.$$

This is a finite representational range, not a replacement for the mathematical symbol ∞.

The source paper's numerical examples based on 7 ± 2 and 11.11 billion should therefore be recorded as historical motivations, not universal constants. [15]

Research programme 7.8 (Finite-resolution number theory). Study triples (K, d, Q) and characterize the tradeoff

$$(|\mathcal{Q}|, \varepsilon_Q, \text{arithmetic cost}, \text{stability}).$$

For structured numerical domains, ask whether an architecture can minimize representation entropy subject to prescribed algebraic and numerical error budgets.

Chapter 8

Finite Approximations of Infinite Mathematical Structures

8.1 What was wrong with the physical-finiteness argument

The Hitchin/Q-learning paper assumes that an information bound forces an infinite mathematical moduli problem to become a finite Markov decision process. The physical-information premise does not entail this mathematical conclusion. A finite algorithm can specify an infinite mathematical object, and an infinite-dimensional Hilbert space can support states of finite entropy. Modern operator-algebraic work on quantum information in infinite systems makes this latter point especially clear. [3, 4]

The original physical-inviability paper likewise identifies a Bekenstein entropy bound with $k_B \log \dim \mathcal{H}$. That equality is not a general entropy identity. It is appropriate only for a finite microcanonical state model in which the relevant accessible states are equiprobable. The safe statement uses the number of operationally distinguishable states in the specified energy and localization sector. Operator-algebraic treatments of quantum field theory also make clear that entropy bounds must not be silently converted into finite Hilbert-space dimension. [3, 4]

Proposition 8.1 (Finite entropy does not imply finite Hilbert dimension)**.** *There exist density operators with finite von Neumann entropy on an infinite-dimensional Hilbert space. On $\ell^2(\mathbb{N})$ define*

$$\rho = \sum_{n=1}^{\infty} 2^{-n}\, |n\rangle\langle n|.$$

Then $\operatorname{Tr}\rho = 1$ *and*

$$S(\rho) = -k_B \operatorname{Tr}(\rho \log \rho) = 2k_B \ln 2 < \infty,$$

while $\ell^2(\mathbb{N})$ is infinite-dimensional.

Proof. The eigenvalues are $p_n = 2^{-n}$, whose sum is 1. Hence

$$S(\rho) = -k_B \sum_{n=1}^{\infty} 2^{-n} \ln(2^{-n}) = k_B \ln 2 \sum_{n=1}^{\infty} n 2^{-n} = 2k_B \ln 2.$$

Thus finite entropy is compatible with an infinite-dimensional Hilbert space. □

This elementary counterexample already breaks the source implication "finite entropy ⇒ finite Hilbert-space dimension," independently of the subtleties of quantum field theory.

8.2 Operational finite distinguishability

Let $\mathcal{S}(E, R, \mathcal{O})$ be a finite family of pairwise perfectly distinguishable states selected by an operational protocol $\mathcal{O}$ for a system confined to a region of radius R and energy at most E, and write

$$\Omega(E, R, \mathcal{O}) := |\mathcal{S}(E, R, \mathcal{O})|.$$

If the relevant entropy is bounded by a Bekenstein-type inequality

$$S \leq \frac{2\pi k_B E R}{\hbar c},$$

and if the accessible distinguishable microstates are modeled microcanonically by

$$S = k_B \ln \Omega,$$

then

$$\Omega \leq \exp\left(\frac{2\pi E R}{\hbar c}\right). \tag{8.1}$$

This is an operational counting statement, not a theorem that every mathematically relevant Hilbert space must be finite-dimensional. The existence and exact scope of Bekenstein-type bounds remain a subject with nontrivial assumptions and formulations; rigorous treatments exist in quantum field theory, including operator-algebraic approaches. [1, 3, 2]

Theorem 8.2 (Conditional categorical distinguishability bound)**.** *Suppose a finite-resource physical implementation has at most Ω pairwise perfectly distinguishable operational states. Suppose further that every categorical level $k = 0, \ldots, N$ is required by the chosen implementation model to contribute at least one new pairwise distinguishable state. Then*

$$N + 1 \leq \Omega.$$

If, under an explicitly additive information model, level k requires at least r_k bits of distinguishable information, then the corresponding information-cost bound is

$$\sum_{k=0}^{N} r_k \leq \log_2 \Omega.$$

Proof. There can be no more than Ω pairwise distinguishable states by assumption. If every level requires a new such state, there must be at least $N + 1$ states. The bit-cost form follows by summing the required independent information budgets. □

The theorem is deliberately conditional. Without the implementation assumption, no bound on abstract categorical height follows.

8.2.1 Physical categorical finiteness as an operational heuristic

The hard theorem above is intentionally narrow. Nevertheless, it suggests a useful research heuristic that is close to the intuition behind the original physical-categorical programme.

Definition 8.3 (Operational depth at resolution δ)**.** Let $\mathcal{P}_0 \subseteq \mathcal{P}_1 \subseteq \cdots$ be a hierarchy of admissible physical or computational probes, let

$$O_n : \mathcal{P}_n \to (Y, d_Y)$$

be the associated observable outputs, and let I_{phys} denote a finite information or resource budget available to the implementation. For $\delta > 0$, define $N_{\text{eff}}(\delta)$ to be the least N (when it exists) such that every output in $\bigcup_{n \geq 0} O_n(\mathcal{P}_n)$ is within δ of some output already attainable at level N.

Conjecture 8.4 (Operational categorical finiteness principle)**.** *Consider a physically realizable hierarchy of probes with finite resource budget I_{phys}. Assume that, at resolution δ, every genuinely new categorical level beyond N consumes at least $r_\delta > 0$ additional distinguishable information in the operational model. Then there is a finite effective depth, and one expects a bound of the form*

$$N_{\text{eff}}(\delta) \lesssim \frac{I_{\text{phys}}}{r_\delta}.$$

The symbol $\lesssim$ is intentional: without a concrete encoding and a proved resource-accounting model, the statement is a scaling heuristic rather than a theorem.

Remark 8.5 (What this conjecture does not say)**.** This principle does *not* say that an abstract ∞-category is itself finite, that all high categorical morphisms disappear, or that finite entropy forces finite Hilbert dimension. It says only that a particular physical implementation may have finite *operationally distinguishable depth* at a declared resolution. Different encodings can have different r_δ, and an implementation can compress many categorical levels into the same observable state. Thus the principle is a rule of thumb, not a hard-and-fast law of nature.

Proposition 8.6 (Exact form under an additive cost hypothesis)**.** *If the implementation satisfies the stronger assumption that each new level k requires a provably independent cost $r_k > 0$ and the total available cost is at most I_{phys}, then any distinguishable realization of levels $0, \ldots, N$ obeys*

$$\sum_{k=0}^{N} r_k \leq I_{\mathrm{phys}}.$$

In particular, if $r_k \geq r_\delta > 0$, then

$$N + 1 \leq \frac{I_{\mathrm{phys}}}{r_\delta}.$$

Proof. Additivity gives the first inequality. The second follows from $(N+1)r_\delta \leq I_{\mathrm{phys}}$. □

The proposition identifies exactly what would turn the heuristic into a theorem: one must supply a physically meaningful information cost per genuinely new categorical level. That is the missing bridge in the original programme.

Remark 8.7 (Bekenstein-scaled rule of thumb)**.** Under the additional microcanonical identification used in (8.1), the available distinguishable information is bounded by

$$I_{\mathrm{phys}} := \log_2 \Omega \leq \frac{2\pi ER}{\hbar c \ln 2}.$$

If a particular implementation empirically or theoretically supports a positive effective cost r_δ bits per genuinely new level at resolution δ, the heuristic becomes the concrete scaling estimate

$$N_{\mathrm{eff}}(\delta) + 1 \ \lesssim \ \frac{2\pi ER}{\hbar c \ln 2 \ r_\delta}.$$

This should be read as an *operational scaling rule*, not as a universal physical law. It can fail when levels are compressed, when the cost per level tends to zero, when the accessible-state model is not microcanonical, or when the physical entropy bound does not apply to the chosen system.

8.3 Finite approximation of a moduli object

Theorem 8.8 (Finite net for compact metric models)**.** *Let (K, d) be compact and let $\varepsilon > 0$. Then there exists a finite set $K_\varepsilon \subset K$ such that every $x \in K$ satisfies*

$$\min_{q \in K_\varepsilon} d(x, q) \leq \varepsilon.$$

Consequently, a finite ε-net and hence a finite geometric state representation can be chosen without invoking physics, provided the computational task has first been reduced to a compact metric model. Preservation of a specific observable, transition rule, or decision problem still requires the corresponding continuity or error-control hypotheses supplied elsewhere in the notebook.

Proof. The open balls $B(x, \varepsilon)$ for $x \in K$ form an open cover of K. Compactness gives a finite subcover, whose centers form K_ε. □

Theorem 8.9 (Propagation of discretization error along a Lipschitz map)**.** *Let $F : K \to K$ be L-Lipschitz and let $Q_\varepsilon : K \to K_\varepsilon$ satisfy*

$$d(Q_\varepsilon x, x) \leq \varepsilon.$$

Define the discretized dynamics $F_\varepsilon = Q_\varepsilon \circ F$. Then, for the exact orbit $x_{n+1} = F(x_n)$ and discretized orbit $\hat{x}_{n+1} = F_\varepsilon(\hat{x}_n)$ with $e_n = d(x_n, \hat{x}_n)$,

$$e_{n+1} \le Le_n + \varepsilon,$$

and hence

$$e_n \le L^n e_0 + \varepsilon \sum_{j=0}^{n-1} L^j.$$

If $L < 1$, then

$$\limsup_{n\to\infty} e_n \le \frac{\varepsilon}{1-L}$$

when $e_0 = 0$.

Proof. By the triangle inequality,

$$d(F(x_n), Q_\varepsilon F(\hat{x}_n)) \le d(F(x_n), F(\hat{x}_n)) + d(F(\hat{x}_n), Q_\varepsilon F(\hat{x}_n)) \le Le_n + \varepsilon.$$

The stated bounds follow by induction. □

Let $\mathcal{M}$ be a mathematical moduli space or stack, and suppose an actual computational task only requires a compact subset $K \subset \mathcal{M}$ and a metric or pseudometric d on a concrete parameterization. Let

$$Q_\varepsilon : K \to K_\varepsilon$$

be a finite representative set with

$$\sup_{x\in K} d(x, Q_\varepsilon x) \le \varepsilon.$$

This is the appropriate place to introduce a finite-state search problem.

Definition 8.10 (δ-commutativity)**.** Given maps $F, G : K \to Y$ into a metric space (Y, d_Y), a numerical diagram is δ-commutative on K if

$$\sup_{x\in K} d_Y(F(x), G(x)) \le \delta.$$

Proposition 8.11 (Error propagation)**.** *Suppose $F = F_2 \circ F_1$ and $G = G_2 \circ G_1$, and suppose*

$$\sup_x d(F_1x, G_1x) \le \eta_1, \qquad \sup_y d(F_2y, G_2y) \le \eta_2,$$

with F_2 L-Lipschitz. Then

$$\sup_x d(Fx, Gx) \le L\eta_1 + \eta_2.$$

Proof. By the triangle inequality,

$$d(F_2F_1x, G_2G_1x) \le d(F_2F_1x, F_2G_1x) + d(F_2G_1x, G_2G_1x),$$

and the two terms are bounded by $L\eta_1$ and η_2 respectively. □

This gives a rigorous route from discretization errors to a provable δ budget.

Corollary 8.12 (Observable finite-state reduction)**.** *Let K be compact and let $O : K \to (Y, d_Y)$ be continuous. Then $O(K)$ is compact and therefore totally bounded. Hence, for every $\delta > 0$, the observable outputs admit a finite δ-net. A finite-state approximation is therefore mathematically justified whenever the computational task depends only on a continuous observable of a compact reduced model, without appealing to a physical entropy argument.*

Proof. Continuous images of compact sets are compact. Every compact metric space is totally bounded. □

8.4 What is required before reinforcement learning

The words "state", "action", and "path" must denote actual mathematical objects before a learning theorem can be invoked. In particular, a Homotopy Type Theory path is not automatically a finite action in a tabular MDP. One must specify a finite encoding of admissible actions and a transition rule that remains inside the finite state set.

8.4.1 A salvageable Hitchin-to-learning heuristic

The strongest defensible version of the original idea is not a theorem saying that every Hitchin moduli problem *is* a finite MDP. It is a conditional approximation architecture: The central heuristic is that learning may become useful once the infinite geometric object has first been reduced to the finite observables actually needed by the computational task.

$$\text{moduli object} \to \text{compact operational sector} \to \text{finite observable net},$$
$$\text{finite action discretization} \to \text{approximate MDP} \to \text{Q-learning}.$$

Definition 8.13 (Admissible geometric control model)**.** An admissible geometric control model consists of a compact metric set (K, d), a finite family of observables $O = (O_1, \dots, O_m)$ with $O_i : K \to Y_i$, a finite-dimensional or otherwise effectively representable action space U, a controlled evolution

$$F : K \times U \to K,$$

and a bounded reward functional $r : K \times U \to [-R_{\max}, R_{\max}]$. The model is *Markov-closed* if the future observable behavior depends on the present only through the chosen finite state representation and action.

Research programme 8.14 (Hitchin-to-MDP reduction)**.** For a specified Hitchin moduli problem, seek a compact operational sector K, a continuous observable map O, a finite ε-net K_ε, a finite action set A_ε, and transition/reward approximations

$$\hat{F}_\varepsilon, \qquad \hat{r}_\varepsilon,$$

with explicit bounds. The target is not exact equality of the moduli problem with the MDP, but a certified statement that decisions made on the finite model approximate decisions for the chosen geometric observable task.

The geometric starting point is not vacuous, but its hypotheses must be stated precisely. For Higgs-bundle moduli over a smooth projective curve of genus at least two, the Hitchin morphism is a proper morphism in the standard semistable settings developed by Hitchin and Nitsure. Properness of a morphism to the Hitchin base is not the same statement as compactness of the total moduli space. Thus the programme should seek a compact *operational sector* rather than silently identify the whole moduli problem with a compact state space. [16, 17]

Conjecture 8.15 (Geometric-to-RL approximation principle)**.** *Suppose a Hitchin-type geometric control problem admits an admissible geometric control model on a compact operational sector K, and suppose:*

(i) K admits finite ε-nets;

(ii) the relevant observables and controlled dynamics are uniformly continuous (Lipschitz is sufficient);

(iii) actions can be discretized with a controlled error;

(iv) the resulting finite representation is approximately Markov-closed;

(v) rewards are bounded and their discretization error is controlled; and

(vi) the induced finite MDP is discounted with $0 \le \gamma < 1$.

Then one should expect a finite MDP whose optimal policy is approximately optimal for the original observable task, with an error bound obtained by summing geometric, action, transition, and reward discretization errors and propagating them through the discounted Bellman operator.

Remark 8.16 (Status of the principle)**.** This is a conjectural *architecture*, not a theorem about Hitchin stacks in general. The individual pieces are standard once their hypotheses hold; the unresolved part is the construction of a useful compact operational sector and a finite Markov representation that faithfully captures the geometric task. In this form the claim is testable: a concrete example either supplies the required objects and error bounds or it does not.

Definition 8.17 (Certified finite search model)**.** A certified discretized search model consists of a finite state set S, finite action sets $A(s)$, a transition kernel $P(\cdot \mid s,a)$, a bounded reward $r(s,a)$, a discount factor $\gamma \in [0,1)$, and an observable error map

$$e : S \times A \to [0,\infty).$$

A successful terminal state additionally satisfies $e(s,a) \le \delta$ for the required δ-criterion.

8.5 Q-learning: what it can and cannot prove

Once a concrete finite state space S, finite action space A, transition kernel, bounded reward function, and discount factor $0 \le \gamma < 1$ have been defined, tabular Q-learning has a classical convergence theorem under repeated sampling and standard step-size conditions. Watkins and Dayan prove almost-sure convergence to optimal action values when all state-action pairs are repeatedly sampled in the discrete setting. [5]

A precise form of the classical conditions is the following. For each state-action pair (s,a), let $\alpha_t(s,a)$ denote the update step used on its successive visits. Assume

$$0 \le \alpha_t(s,a) < 1, \qquad \sum_t \alpha_t(s,a) = \infty, \qquad \sum_t \alpha_t(s,a)^2 < \infty,$$

and assume every state-action pair is visited infinitely often. Together with finite state and action sets, bounded rewards, and a discounted objective, these conditions give the standard almost-sure convergence result. [5]

The original Hitchin paper's claimed convergence theorem therefore needs to be split into two statements:

i. *algorithmic theorem*: Q-learning converges on the specified finite MDP under standard assumptions;
ii. *geometric theorem*: the finite MDP approximates the intended mathematical moduli problem to a specified error.

Only together do these imply a useful approximation result.

Proposition 8.18 (Value stability under reward perturbation)**.** *Let two discounted finite MDPs share the same finite state and action sets, transition kernel P, and discount factor $0 \le \gamma < 1$, but have bounded rewards r and r' satisfying*

$$\sup_{s,a} |r(s,a) - r'(s,a)| \le \varepsilon_r.$$

Then their optimal value functions satisfy

$$\sup_s |V^*(s) - V'^*(s)| \le \frac{\varepsilon_r}{1-\gamma}.$$

Proof. Let T and T' be the optimal Bellman operators. For every bounded value function V,

$$\|TV - T'V\|_\infty \le \varepsilon_r,$$

and both operators are γ-contractions in the sup norm. Therefore

$$\|V^* - V'^*\|_\infty \le \varepsilon_r + \gamma \|V^* - V'^*\|_\infty,$$

which gives the stated inequality after rearranging. □

Proposition 8.19 (Value stability under transition and reward perturbations)**.** *Let two discounted finite MDPs have the same finite state and action sets and discount factor $0 \le \gamma < 1$. Let rewards satisfy*

$$\sup_{s,a} |r(s,a) - r'(s,a)| \le \varepsilon_r,$$

and suppose the transition kernels satisfy

$$\sup_{s,a} \|P(\cdot \mid s,a) - P'(\cdot \mid s,a)\|_{\mathrm{TV}} \le \varepsilon_P.$$

If rewards are bounded by $R_{\max}$ in absolute value, then

$$\left\|V^* - V'^*\right\|_\infty \leq \frac{\varepsilon_r}{1-\gamma} + \frac{2\gamma R_{\max}\varepsilon_P}{(1-\gamma)^2}.$$

Proof. For any bounded V with $\|V\|_\infty \leq R_{\max}/(1-\gamma)$,

$$|\mathbb{E}_P V - \mathbb{E}_{P'} V| \leq 2\,\|V\|_\infty\,\|P - P'\|_{\mathrm{TV}}\,.$$

Hence the Bellman operators differ by at most

$$\varepsilon_r + \frac{2\gamma R_{\max}\varepsilon_P}{1-\gamma}.$$

Using the contraction argument from the previous proposition and dividing by $1-\gamma$ gives the result. □

This estimate supplies a useful quantitative target for the Hitchin programme: it separates local geometric/reward approximation error from the dynamic amplification caused by discounting.

Proposition 8.20 (Greedy-policy transfer bound)**.** *Suppose two finite discounted MDPs share the same finite state and action sets and the same transition kernel P, and have optimal action-value functions Q^* and $\widehat{Q}^*$ satisfying $\|Q^* - \widehat{Q}^*\|_\infty \leq \delta_Q$. Let π be greedy with respect to $\widehat{Q}^*$. Then*

$$0 \leq V^*(s) - V^\pi(s) \leq \frac{2\delta_Q}{1-\gamma} \tag{8.2}$$

for every state s.

Proof. At each state, greediness gives $\widehat{Q}^*(s,\pi(s)) \geq \widehat{Q}^*(s,a^*)$ for an optimal action a^*. The uniform error bound therefore implies $Q^*(s,\pi(s)) \geq V^*(s) - 2\delta_Q$. Writing $d(s) = V^*(s) - V^\pi(s)$ and using the definition of Q^* then yields $d(s) \leq 2\delta_Q + \gamma\sum_{s'} P(s' \mid s,\pi(s))d(s')$. Taking the supremum gives $\|d\|_\infty \leq 2\delta_Q + \gamma\|d\|_\infty$, which rearranges to the stated bound. □

This lemma does not solve the Hitchin approximation problem because changing the state representation generally changes the state and transition spaces themselves. It does, however, close one important layer of the logic: once the same finite MDP has been constructed, a controlled reward approximation gives a controlled decision-value error.

Research programme 8.21 (Categorical approximation under resource constraints)**.** Given a concrete moduli problem, construct K_ε, a finite state representation, a metric on the relevant observables, and an explicit error theorem relating discrete paths to continuous or categorical paths. Only after this should learning or combinatorial search be compared experimentally with deterministic numerical methods.

Chapter 9

Operational Depth as a Hitting-Time Theory

9.1 Discrete arithmetic contexts

The source paper defines an arithmetic context $(S, \oplus, O)$ and an inverse operation. For additive contexts, the reduction rule is subtraction; for multiplicative contexts, it is division. The examples are correct on their natural domains, but exact discrete termination requires arithmetic compatibility.

Definition 9.1 (Termination depth)**.** Let $F : S \to S$ and target set $A \subseteq S$. The discrete termination depth is

$$\tau_A(x) := \inf\{n \in \mathbb{N} : F^n(x) \in A\},$$

with $\tau_A(x) = \infty$ if the set is empty.

For subtraction by a fixed $b \in \mathbb{N}_{>0}$, define the admissible state set

$$S_b = \{0, b, 2b, 3b, \ldots\}, \qquad F(x) = x - b.$$

Then $F : S_b \setminus \{0\} \to S_b$ and finite termination at 0 occurs for every $x \in S_b$, with

$$\tau_{\{0\}}(x) = x/b.$$

Equivalently, on the ambient state space $\mathbb{N}$, finite termination by repeated subtraction of b occurs exactly for inputs divisible by b, after restricting the iteration to admissible states. Thus division appears as a special case of hitting time, but the state-domain condition is part of the definition.

Remark 9.2 (Discrete versus continuous division)**.** The equation $T(a, b) = a/b$ is not a statement about an integer iteration count on all of $\mathbb{R}$. In the discrete model the count belongs to $\mathbb{N}$, so termination requires $a/b \in \mathbb{N}$. The real-valued quotient a/b belongs to the continuous flow formulation instead.

For positive multiplicative states, let

$$F_b(x) = x/b, \qquad b > 0,\ b \neq 1.$$

A forward discrete hit of 1 occurs exactly when there exists $n \in \mathbb{N}$ with $x = b^n$ and, necessarily, the direction of the orbit places 1 ahead of x. In that case

$$\tau_{\{1\}}(x) = n = \log_b x.$$

The sign of $\log_b x$ must therefore be compatible with forward time; otherwise the displayed logarithm describes a backward or non-admissible hitting time.

9.2 Symmetric termination and the square root

The source paper defines a symmetric root by requiring the operand and the termination depth to coincide. In the continuous extension this gives a clean characterization.

Proposition 9.3 (Symmetric square-root characterization)**.** *For $a > 0$, the unique positive x satisfying*

$$\frac{a}{x} = x$$

is $x = \sqrt{a}$. Equivalently, the continuous additive hitting time of a under constant-rate subtraction by x equals the amount x subtracted precisely when $x = \sqrt{a}$.

Proof. The equation $a/x = x$ is equivalent to $x^2 = a$, whose unique positive solution is $\sqrt{a}$. □

The novelty is not a new definition of the square root in mathematics, but an operational characterization of it.

9.3 Continuous extension

For additive flow

$$\dot{y} = -b, \qquad y(0) = a,$$

we obtain

$$y(t) = a - bt, \qquad \tau_0 = \frac{a}{b}.$$

For multiplicative flow

$$\dot{y} = -(\ln b)y, \qquad y(0) = a,$$

we get

$$y(t) = ab^{-t}.$$

For a forward hit of the target 1, assume either $a > 1$ and $b > 1$, or $0 < a < 1$ and $0 < b < 1$. Then

$$\tau_1 = \frac{\ln a}{\ln b} = \log_b a \geq 0.$$

This is a genuine continuous hitting-time interpretation under the stated forward-time conditions.

Theorem 9.4 (Logarithmic conjugacy of multiplicative reduction)**.** *On $(0, \infty)$ let $F_b(x) = x/b$ with $b > 0$, $b \neq 1$, and let $L(x) = \ln x$. Then*

$$L \circ F_b = G_b \circ L, \qquad G_b(y) = y - \ln b.$$

Thus multiplicative reduction is conjugate under the logarithm to additive translation. Discrete and continuous multiplicative hitting times are therefore additive hitting times in logarithmic coordinates.

Proof. For every $x > 0$,

$$L(F_b(x)) = \ln(x/b) = \ln x - \ln b = G_b(L(x)).$$

The hitting-time correspondence follows because L is a bijection from $(0, \infty)$ to $\mathbb{R}$. □

9.4 General dynamical formulation

The same idea applies well beyond arithmetic. For a discrete map F and target A,

$$\tau_A(x) = \inf\{n : F^n(x) \in A\},$$

while for a continuous flow Φ_t,

$$\tau_A(x) = \inf\{t \geq 0 : \Phi_t(x) \in A\}.$$

Hitting and return times are a developed topic in dynamical systems, including distributional and asymptotic results. [6, 7]

Research programme 9.5 (Operational depth invariant)**.** For a class of computational maps F and target sets A, investigate the geometry and complexity of τ_A. Questions include finiteness criteria, continuity or semicontinuity in parameters, expected hitting times under random initial states, and relations between operational depth and algorithmic cost.

Audit conclusion

Operational Depth survives strongly after one correction: exact discrete division/logarithms require orbit compatibility; the continuous version and the general hitting-time formulation are mathematically clean.

Chapter 10

Epistemic Saturation and Decay

10.1 The correct semantic object

The source paper describes pointed Kripke models and proposes a knowledge endofunctor, but the functorial description is not needed for the core theorem and is under-specified as written. We instead work directly with a finite multi-agent Kripke model

$$M = (W, \{R_a\}_{a\in A}, V),$$

where W is finite, $R_a \subseteq W \times W$ is the accessibility relation for agent a, and V is the valuation of atomic propositions.

Mathematical boundary

The source paper's notation $K : \mathcal{E} \to \mathcal{E}$ is not, by itself, an endofunctor. To call "knows that" a functor one must specify both its action on pointed Kripke objects and its action on p-morphisms, and then verify preservation of identities and composition. Since the source does not supply those data, the categorical iteration $K^n(X)$ is not used as the foundation of the stabilization theorem below.

10.2 Depth-n Behavioral Types

Assume that the set of agents A and the set of atomic propositions used in the initial valuation are finite. Define the atomic type

$$\tau_0(w) = V(w)|_{\mathsf{At}},$$

where At is the chosen finite or explicitly represented set of atomic propositions used to initialize the refinement. Recursively define

$$\tau_{n+1}(w) = \left(\tau_0(w), (\{\tau_n(v) : wR_av\})_{a\in A}\right),$$

where, for each agent a, the successor information is represented by the set of τ_n-classes reached by R_a-successors. Multiplicity is not relevant for ordinary Kripke bisimulation; if multiplicity is intended to matter, a different notion must be specified. Two worlds are n-equivalent when their types agree.

Let Π_n be the partition of W induced by τ_n. Then

$$\Pi_{n+1} \preceq \Pi_n$$

meaning that Π_{n+1} is a refinement of Π_n.

Theorem 10.1 (Finite epistemic stabilization)**.** *If W is finite, there exists $N \leq |W| - 1$ such that*

$$\Pi_N = \Pi_{N+1} = \Pi_{N+2} = \cdots .$$

Proof. Each strict refinement of a partition of W increases the number of blocks by at least one. The number of blocks is at most $|W|$. Therefore there can be at most $|W| - 1$ strict refinements. Once a step produces no refinement, the deterministic refinement rule produces no later refinement. □

This is the precise core behind the source intuition. It is closely related to finite-depth modal equivalence and bisimulation/partition-refinement methods. [8, 9]

Theorem 10.2 (Stabilized partition is a bisimulation quotient)**.** *At the first index N for which $\Pi_N = \Pi_{N+1}$, the induced equivalence relation is stable under atomic valuation and successor-block matching. Hence it is a bisimulation equivalence for the finite Kripke frame (with one relation for each agent), and the quotient by Π_N preserves all formulas of the basic multi-agent modal language generated by the chosen finite atomic set At.*

Proof. Atomic agreement is built into τ_0. Equality of τ_{N+1} means that equivalent worlds have exactly the same sets of Π_N-blocks among their R_a-successors for each agent a. Therefore the forth and back conditions for a bisimulation are satisfied. Induction on modal formula complexity then gives preservation of truth. □

10.3 Epistemic saturation depth

Definition 10.3. For a fixed model and language, define

$$N_{\text{sat}}(M) := \min\{N : \Pi_N = \Pi_{N+1}\}.$$

This is a real invariant of the chosen finite semantic structure.

10.4 Why the proposed logarithmic decay law fails

The source paper proposes approximately

$$N \approx \frac{\log |W|}{\log b}$$

and a decay rate proportional to its reciprocal. No such formula follows from the stabilization proof. In particular, graph diameter, branching factor, label structure, relation geometry, and multi-agent composition can affect refinement behavior in ways that are not determined by $(|W|, b)$ alone.

Therefore the logarithmic law is discarded as a theorem. A replacement must be empirical or derived for a restricted graph family.

Example 10.4 (Same state count, different saturation depths)**.** A finite Kripke frame may have $N_{\text{sat}} = 0$ when the initial atomic partition is already stable under successor-block signatures. For contrast, take a finite transition system with one relation on worlds $w_0, \dots, w_{m-1}$ and

$$w_0 R w_1 R \cdots R w_{m-1}, \qquad R(w_{m-1}) = \varnothing,$$

with no distinguishing atomic propositions. Each refinement round separates one additional position in the chain, so $N_{\text{sat}} = m - 1$. This proves sharpness of the $|W| - 1$ bound for general finite Kripke frames. For genuine epistemic models in the usual S5 sense, the same finite stabilization theorem holds, but sharpness is a separate structural question and is not asserted here.

10.5 A rigorous decay functional

The word "decay" can be made quantitative by selecting a complexity or decision functional J on partitions. For example,

$$D_n := J(\Pi_n) - J(\Pi_{n+1}) \geq 0$$

whenever J is chosen to decrease under refinement. Alternatively, let U_n denote the expected optimal utility available to an agent using only depth-n information, and define

$$\Delta_n := U_{n+1} - U_n.$$

Then "epistemic decay" means Δ_n becomes small, while "saturation" means exact semantic equality of the relevant partitions.

Definition 10.5 (Decision-relative epistemic horizon)**.** For tolerance $\delta > 0$, define

$$N_\delta := \min\{n : \Delta_k \le \delta \text{ for all } k \ge n\}$$

when such an index exists.

This separates a mathematical stabilization theorem from an application-dependent utility threshold.

10.6 Cybersecurity and Needham–Schroeder

The source case study correctly identifies the classical Lowe attack as an epistemic failure involving a responder identity not being bound into the protocol message. Lowe's 1995 paper gives the attack itself; the 1996 TACAS work reports formal modeling and verification of the protocol and its repair. [18, 10]

The safe Notebook-III statement is therefore not "the protocol is fixed because $N = 2$" as a universal theorem. Instead, construct a finite protocol model, define the exact proposition language and observation model, compute Π_n, and test whether the repaired protocol has a smaller N_{sat} or a smaller decision-relative horizon.

Research programme 10.6 (Epistemic security horizon)**.** For finite protocol models, measure N_{sat}, N_δ, and the marginal decision value Δ_n. Compare vulnerable and repaired protocols under identical state spaces and observation models. Determine which graph-theoretic and protocol-structural parameters control these quantities.

Chapter 11

Physical Versus Epistemic Finiteness

11.1 Separating three notions of finiteness

The source physical-finiteness paper conflates three statements:

$$\text{finite resource capacity,}$$
$$\text{finite observable state count,}$$
$$\text{finite abstract mathematical dimension.}$$

The first can imply the second under specified physics and measurement assumptions. Neither generally implies the third.

11.2 Operational categorical depth

Let $\mathcal{C}$ be an abstract higher-categorical structure and let

$$\mathcal{O} : \mathcal{C} \to \mathcal{D}$$

be an operational observation map into a finite or compactly approximated observable system. Define depth-n observational equivalence by agreement on all probes of categorical level at most n.

Definition 11.1 (Operational categorical saturation)**.** When such an index exists, the operational depth at tolerance δ is the least N for which every admissible level-n probe with $n > N$ is observationally approximated, within δ, by some probe available at a level $k \leq N$ in the specified implementation model.

This is the corrected descendant of the original "finite categorical order" idea. It makes no claim that the abstract ∞-category itself ceases to exist.

Definition 11.2 (Observable level sets)**.** Let $\mathcal{P}_n$ be the admissible probes available at categorical level n, let

$$\mathcal{O}_n : \mathcal{P}_n \to (Y, d_Y)$$

be their observable outputs, and define the cumulative observable sets

$$A_n := \bigcup_{k=0}^{n} \mathcal{O}_k(\mathcal{P}_k), \qquad A_\infty := \bigcup_{n \geq 0} A_n.$$

Then $A_n \subseteq A_{n+1}$ automatically.

Theorem 11.3 (Finite observational saturation from total boundedness)**.** *Suppose A_∞ is totally bounded in (Y, d_Y). Then for every $\delta > 0$ there exists N such that*

$$A_\infty \subseteq \bigcup_{y \in A_N} B(y, \delta).$$

Equivalently, every observable output from every depth is δ-approximated by an output already available by one finite level N.

Proof. Total boundedness gives a finite $\delta/3$-net $F \subset Y$ for A_∞. For each $f \in F$ that is within $\delta/3$ of some point of A_∞, choose one such point $a_f \in A_\infty$. Since F is finite and the sets A_n are nested, there is a common index N containing all the chosen a_f. Every $a \in A_\infty$ is within $\delta/3$ of some $f \in F$, and that f is within $\delta/3$ of $a_f \in A_N$. Therefore $d_Y(a, a_f) < 2\delta/3 < \delta$. □

This theorem is purely mathematical. A finite physical resource budget may be used to motivate or establish total boundedness of an observable model, but that physical implication is an additional hypothesis rather than a consequence of categorical language alone. Total boundedness guarantees existence of a finite effective depth at tolerance δ, but by itself gives no explicit numerical bound on that depth; an explicit bound requires information about how quickly the level sets exhaust a finite net.

11.3 Relation to epistemic saturation

The structural parallel is now precise:

$$N_{\mathrm{epi}}(M, \mathcal{L}) \qquad \text{versus} \qquad N_{\mathrm{eff}}(\mathcal{C}, B, \delta, \mathcal{O}).$$

The first is purely semantic and finite-model theoretic; the second is physical/operational. A future theorem relating them would require an explicit encoding functor and error model.

Mathematical boundary

It is not currently justified to claim that physics proves all abstract ∞-categories physically impossible, nor that every finite physical system has a finite-dimensional Hilbert space solely because a Bekenstein-type entropy bound is finite. The correct claim concerns finitely many *accessible and distinguishable states or probes under a specified operational model.*

Chapter 12

The God Monad as a Context Monad

12.1 The exact mathematics

The source paper defines a two-element context set

$$G = \{0, 1\}$$

and the endofunctor

$$T_G(X) = X^G.$$

This is the standard Reader/Environment monad. [11]

For $f : X \to Y$,

$$T_G(f)(h) = f \circ h.$$

The unit is

$$\eta_X(x)(g) = x,$$

and the multiplication is

$$\mu_X(k)(g) = k(g)(g).$$

Theorem 12.1 (Reader monad laws)**.** *For every set G, (T_G, η, μ) is a monad on* **Set***.*

Proof. Functoriality is immediate from composition. For left unit, given $h : G \to X$,

$$\mu_X(T_G\eta_X(h))(g) = \eta_X(h(g))(g) = h(g).$$

For right unit,

$$\mu_X(\eta_{T_G(X)}(h))(g) = h(g).$$

For associativity, if $K : G \to (G \to (G \to X))$, then both sides evaluate at g to

$$K(g)(g)(g).$$

Thus all three monad laws hold. □

12.2 What the name does and does not mean

Choosing $G = \{\text{exists}, \text{does not exist}\}$ is a philosophical labeling of an ordinary environment monad. The mathematics does not prove either existence or non-existence of the corresponding proposition.

What it does provide is a formal context in which a computation may depend on an unresolved world variable. More generally,

$$G = \{\text{models}, \text{worlds}, \text{hypotheses}, \text{environments}\}$$

and X^G is a context-indexed family of outputs.

12.3 Context collapse as a natural transformation

For each fixed $g_0 \in G$, evaluation

$$\mathrm{ev}_{g_0} : X^G \to X, \qquad h \mapsto h(g_0)$$

is natural in X. It is a monad algebra, expressing the selection of one context. Thus the richer contextual object can be collapsed to a single-world model when the context has been fixed.

12.4 The philosophical boundary

The source paper's "argument from structural comprehensiveness" does not follow from the monad laws. Greater expressive capacity is a mathematical property; rational preference for one epistemic stance over another requires an additional decision or evidential criterion.

Salvage

The clean mathematical contribution is therefore the general *context monad* viewpoint. This is standard monadic structure in the categorical semantics of computation. [11, 12] The theological origin remains historically important, but the formalism is broader than the original application.

Chapter 13

Cross-Program Synthesis

13.1 Five streams

After correction and integration, the source programmes organize into five mathematical streams.

Stream A: iteration and stabilization

$$\begin{gathered}\text{root limits} \longrightarrow \text{principal-branch dynamics} \longrightarrow \text{recursive coordinates}\\ \longrightarrow \text{operational depth} \longrightarrow \text{hitting times} \longrightarrow \text{epistemic refinement}\\ \longrightarrow \text{saturation.}\end{gathered}$$

The common object is repeated transformation, but the limiting notion changes from ordinary convergence to periodic attraction, finite hitting, or finite stabilization.

Stream B: finite representation

$$\begin{gathered}\text{practical zero/infinity} \longrightarrow \text{quantization} \longrightarrow \text{covering numbers}\\ \longrightarrow \text{operational distinguishability} \longrightarrow \text{approximate moduli computation.}\end{gathered}$$

The common question is how much structure can be represented or distinguished at finite resolution and resource budget.

Stream C: geometric certification

$$\begin{gathered}\text{ellipse heuristic} \longrightarrow \text{exact anisotropic ellipsoid bound}\\ \longrightarrow \text{rotated ellipsoid} \longrightarrow \text{general exclusion certificates.}\end{gathered}$$

This stream is the most directly algorithmic and the cleanest source of exact geometric certificates.

Stream D: equality, quotient, and shape

$$\begin{gathered}\text{located geometric objects} \longrightarrow \text{equivalence relations}\\ \longrightarrow \text{set quotients} \longrightarrow \text{normalized shape spaces}\\ \longrightarrow \text{invariant properties.}\end{gathered}$$

The triangle chapter makes explicit a pattern that recurs elsewhere in the volume: a raw representation carries more structure than the intended invariant, and a mathematically controlled quotient separates the two without confusing strict identity with equivalence.

Stream E: contextual computation and operational semantics

$$\text{compact observable sectors} \longrightarrow \text{finite approximations}$$
$$\longrightarrow \text{controlled learning} \longrightarrow \text{epistemic and categorical probes}$$
$$\longrightarrow \text{context-indexed computation.}$$

The common principle is that an abstract structure becomes operational only after the state, observation, error, and stopping models are made explicit.

13.2 A candidate Pulickal operational principle

A useful synthesis, stated cautiously, is:

> An abstract computational or geometric structure becomes mathematically operational when its state space, transformation or equivalence rule, observation or error model, resource budget, and stopping criterion are all explicitly specified.

This principle is not itself a theorem. It is an editorial and research design rule extracted from the corrected mathematics. The quotient chapters add one further condition: whenever the intended notion of sameness is coarser than strict equality, the equivalence relation and its quotient must be defined before invariant reasoning is claimed.

13.3 Relation to the preceding notebooks

Notebook III is intentionally self-contained. Some themes are continuations of work developed earlier, but no theorem in this volume relies on an unstated result from Notebook I or Notebook II. The present volume records only the parts of those themes needed for the integrated programmes treated here, and it explicitly separates historical continuity from mathematical dependence.

Chapter 14

Priority Research Problems

P1. General ellipsoid certificate. Prove, classify, and implement the exact principal-vertex farthest-distance formula in arbitrary dimension, including degeneracies and repeated largest axes.

P2. Certificate completeness gap. Characterize the unresolved region $\{p : C(p) \le 0\} \setminus E$ and optimize anchor families.

P3. Root-limit structure. Classify convergence, subsequence behaviour, and branch dependence for principal and multivalued root families beyond the scalar and square-root cases treated here.

P4. Recursive-root algebra. Characterize kernels and compatible quotient algebras for finite formal coordinate spaces generated by recursive complex roots.

P5. Finite-resolution number spaces. Compare lattice, floating-point, interval, and adaptive representations using $(|\mathcal{Q}|, \varepsilon, \mathrm{cost}, \mathrm{stability})$.

P6. Triangle-shape quotients. Formalize the effectiveness of congruence and similarity quotients in a proof assistant and compare set-quotient, groupoid, and stack-level representations where automorphisms matter.

P7. Operational depth. Classify maps for which hitting-time depth is finite, continuous, or asymptotically analyzable.

P8. Epistemic saturation. Determine sharp bounds on N_{sat} for important graph families and multi-agent relation classes.

P9. Decision-relative decay. Relate semantic partition stabilization to actual changes in optimal decisions under explicit utility models.

P10. Physical distinguishability. Develop rigorous operational bounds linking resource constraints to the number of distinguishable computational probes without claiming an ontological cutoff for abstract mathematics.

P11. Moduli approximation. Build one concrete finite-dimensional moduli problem for which discretization error and search error can both be bounded independently.

P12. Context monads. Generalize the two-world Reader monad into a family of context-indexed reasoning calculi and study interactions with probability, nondeterminism, or state monads only when coherence conditions are proved.

Chapter 15

Final Verification Ledger

Programme	Original status	Verified replacement
Recursive imaginary topology	Mixed	Formal vector space + realization map + exact branch dynamics; no new-number-system claim.
Approaching Unity	Mixed	Domain- and branch-sensitive root limits, Hausdorff convergence of full roots, and exact principal-branch attractor.
Homotopy Type Theory and geometric equality	Standalone preprint	Located triangle type + proposition-valued congruence/similarity + effective set quotients + normalized shape realization.
Ellipse anomaly work	Heuristic/overclaimed	Historical precursor to one-sided geometric certificates.
Ellipsoid paper	Partly correct	Exact farthest-distance theorem; general $2a_j$ rule restricted.
Number of Numbers	Mathematically invalid counting model	Finite-resolution representation, covering numbers, operational zero/range.
Hitchin/Q-learning	Speculative theorem	Conditional approximation architecture + finite-MDP Q-learning theorem + quantitative reward/transition stability.
Operational Depth	Strong intuition	Hitting-time invariant with arithmetic and continuous special cases.
Epistemic Decay	Mixed	Finite partition stabilization + decision-relative decay functional.
Physical categorical finiteness	Invalid universal theorem	Conditional distinguishability theorem + operational finiteness conjecture/rule of thumb.
God Monad	Correct core, overextended interpretation	General Reader/context monad.

Audit conclusion

The most important verification result is two-sided. Several original claims fail in their stated form, but the failures expose precise mathematical replacements rather than dead ends. The strongest results now include an exact ellipsoid extremum formula, a compactness-based finite-net theorem, a Lipschitz error-propagation bound, finite epistemic stabilization to a bisimulation quotient, and a total-boundedness theorem for finite observational categorical depth. The remaining physical and application-specific bridges are explicitly conditional or conjectural. The manuscript nevertheless preserves a stronger research hypothesis: finite operational budgets plausibly induce finite effective observational depth, and compact observable sectors plausibly admit finite controlled-learning reductions. These are research heuristics with explicit failure modes, not hard laws. No single theorem is claimed to unify all nine programmes.

Appendix A

General Principal-Root Dynamics

A.1 Exact polar recursion

Fix an integer $r \geq 2$ and define the principal map

$$F_r(z) = \mathrm{Root}_r^{\mathrm{pr}}(-z).$$

For a nonzero point write

$$z_n = \rho_n e^{\mathrm{i}\theta_n}, \qquad \rho_n > 0, \qquad \theta_n \in (-\pi, \pi].$$

The principal radial recurrence is

$$\rho_{n+1} = \rho_n^{1/r},$$

and hence

$$\rho_n = \rho_0^{1/r^n} = \exp\left(\frac{\log \rho_0}{r^n}\right) \longrightarrow 1.$$

The angular recurrence is forced by the principal-value convention:

$$\theta_{n+1} = \begin{cases} (\theta_n + \pi)/r, & -\pi < \theta_n \leq 0, \\ (\theta_n - \pi)/r, & 0 < \theta_n \leq \pi. \end{cases}$$

Define $\alpha_n = |\theta_n|$. Then

$$\alpha_{n+1} = \frac{\pi - \alpha_n}{r}.$$

The unique fixed point is

$$\alpha_* = \frac{\pi}{r+1}.$$

Theorem A.1 (Exact angular convergence)**.** *For every nonzero initial point, the principal angular magnitudes satisfy*

$$\alpha_n - \alpha_* = \left(-\frac{1}{r}\right)^n (\alpha_0 - \alpha_*).$$

Consequently $\alpha_n \to \alpha_*$ at rate r^{-n}, while the sign alternates after at most finitely many endpoint steps. Hence the principal orbit has the two-cycle

$$e^{\mathrm{i}\pi/(r+1)} \longleftrightarrow e^{-\mathrm{i}\pi/(r+1)}$$

as its nonzero angular limit set.

Proof. Subtract α_* from the affine recurrence:

$$\alpha_{n+1} - \alpha_* = \frac{\pi - \alpha_n}{r} - \frac{\pi}{r+1} = -\frac{1}{r}(\alpha_n - \alpha_*).$$

Iteration gives the displayed formula. The radial component converges independently to one. Away from the endpoints $\theta = 0$ and $\theta = \pi$, the branch rule changes the sign at every step. If an orbit meets an endpoint, one or two further iterations move it into the alternating regime. The limiting positive and negative angles are therefore $\pm\alpha_*$. $\square$

A.2 Exact moving-cycle error

Let $\varepsilon_n \in \{-1, +1\}$ be the sign of θ_n and define

$$\zeta_n = e^{\mathrm{i}\varepsilon_n \alpha_*}.$$

Then

$$|z_n - \zeta_n|^2 = (\rho_n - 1)^2 + 2\rho_n\bigl(1 - \cos(\alpha_n - \alpha_*)\bigr).$$

Since

$$\rho_n - 1 = \frac{\log \rho_0}{r^n} + O(r^{-2n}), \qquad \alpha_n - \alpha_* = \left(-\frac{1}{r}\right)^n (\alpha_0 - \alpha_*),$$

we obtain

$$|z_n - \zeta_n| = \frac{1}{r^n}\sqrt{(\log \rho_0)^2 + (\alpha_0 - \alpha_*)^2} + O(r^{-2n}).$$

Thus the radial and angular errors combine to first order through a Euclidean norm.

A.3 Branch discontinuity

For every $x > 0$, the two one-sided limits of $F_r(z)$ as $z \to x$ from the upper and lower half-planes differ by the two principal roots of $-x$ with arguments $\pm\pi/r$. Hence the principal map is analytic on each slit branch domain but is not a globally continuous self-map of the plane. Any use of classical Julia-set language must therefore specify an additional branch-sensitive dynamical definition.

Appendix B

Ellipsoid Extremum and Certificate Details

B.1 Reduction to one scalar variable

Let

$$E = \left\{ x \in \mathbb{R}^d : \sum_{i=1}^{d} \frac{x_i^2}{a_i^2} \le 1 \right\}, \qquad a_i > 0,$$

and fix $v_j^- = -a_j e_j$. Put

$$A = \max_i a_i^2, \qquad B = a_j^2.$$

After writing $x_i = a_i y_i$ and $t = y_j$, the squared distance becomes

$$\|x - v_j^-\|_2^2 = B + \sum_i a_i^2 y_i^2 + 2Bt.$$

For fixed t, the remaining quadratic mass is maximized on a direction whose squared semi-axis is A:

$$\sum_i a_i^2 y_i^2 \le Bt^2 + A(1 - t^2).$$

Therefore

$$\|x - v_j^-\|_2^2 \le q(t) = A + B + (B - A)t^2 + 2Bt.$$

If $A \le 2B$, the maximum occurs at $t = 1$, giving $4B$. If $A > 2B$, the stationary point

$$t_* = \frac{B}{A - B}$$

lies in $(0, 1)$ and gives

$$q(t_*) = \frac{A^2}{A - B}.$$

Theorem B.1 (Equality cases)**.** *The exact formula*

$$M_j^2 = \begin{cases} 4B, & A \le 2B, \\ A^2/(A - B), & A > 2B \end{cases}$$

is attained as follows.

If $A \le 2B$, the point $x = +a_j e_j$ is a maximizer.

If $A > 2B$, let U be the subspace spanned by principal axes whose squared semi-axis is A. Any maximizer has $y_j = B/(A - B)$ and its remaining normalized mass lies in U, with total squared norm $1 - t_*^2$.

Proof. The upper bound above is attained whenever the remaining y-mass is supported entirely on an axis or maximal-axis subspace with squared semi-axis A. In the first regime q is maximized at the endpoint $t = 1$. In the second regime q is strictly concave and has the unique stationary point t_* in $(0, 1)$, so every maximizer must realize equality in the quadratic estimate. This gives exactly the stated equality cases. □

Corollary B.2 (Maximizer geometry in the anisotropic regime)**.** *When $A > 2B$, every maximizer lies in the affine subspace generated by the negative e_j-vertex coordinate and the eigenspace of the largest squared semi-axis. If the largest squared semi-axis is simple, there are exactly two maximizers, corresponding to the two signs of the maximal-axis component. If the largest axis has multiplicity greater than one, the maximizers form a sphere inside the corresponding maximal-axis subspace.*

B.2 Rotational invariance

For

$$E_{\mu,\Sigma} = \{x : (x-\mu)^T \Sigma^{-1} (x-\mu) \le 1\},$$

let

$$\Sigma = Q \operatorname{diag}(a_1^2, \dots, a_d^2)\, Q^T$$

with Q orthogonal. Then

$$y = Q^T (x - \mu)$$

preserves Euclidean distances and converts the rotated ellipsoid into principal-axis coordinates. The extremum theorem therefore transports exactly under translation and orthogonal rotation.

B.3 Soundness versus completeness

For anchor points $v_1, \dots, v_m$ with certified radii M_j, define

$$C(p) = \max_j \bigl(\|p - v_j\|_2 - M_j \bigr).$$

Then

$$C(p) > 0 \Longrightarrow p \notin E.$$

The implication is one-sided. The unresolved set

$$\left(\bigcap_j \overline{B}(v_j, M_j) \right) \setminus E$$

may be nonempty, so a complete classifier requires additional structure.

Appendix C

Finite-Resolution Approximation: Further Bounds

C.1 Covering and packing

Let (K, d) be a compact metric space. An ε-net is a finite set $F \subseteq K$ such that every point of K lies within ε of some point of F. The minimal cardinality is the covering number $N(K, \varepsilon, d)$.

If $P \subset K$ is maximal ε-separated, then P is an ε-net. Conversely, packing numbers provide lower bounds for covering numbers at comparable scales. In Euclidean spaces this produces the familiar volume scaling and makes the dimension dependence explicit. [13]

For a Euclidean ball B_R^d there are constants depending only on d such that

$$c_d(R/\varepsilon)^d \leq N(B_R^d, \varepsilon) \leq C_d(1 + R/\varepsilon)^d.$$

C.2 Error growth under repeated discretization

Let $F : K \to K$ be L-Lipschitz and let Q_ε satisfy

$$d(Q_\varepsilon x, x) \leq \varepsilon.$$

Then the error e_n between exact and discretized trajectories obeys

$$e_{n+1} \leq L e_n + \varepsilon.$$

If $L < 1$, then

$$\limsup_{n\to\infty} e_n \leq \frac{\varepsilon}{1 - L}.$$

If $L = 1$, the same recurrence yields at most linear accumulation, while for $L > 1$ it does not give a uniform long-time bound. This is why discretization and stability must be treated as separate hypotheses.

C.3 Observable reduction

If K is compact and $O : K \to Y$ is continuous, then $O(K)$ is compact and hence totally bounded. Therefore every positive tolerance admits a finite observable representation. The underlying mathematical model remains what it was; the finite object is a representation of the observables relevant to the stated task.

Appendix D

Finite MDP Approximation and Learning Guarantees

D.1 Bellman contraction

For a finite discounted MDP with reward bounded by $R_{\max}$ and discount factor $0 \leq \gamma < 1$, the optimal Bellman operator satisfies

$$\|TV - TW\|_\infty \leq \gamma \|V - W\|_\infty.$$

Hence T has a unique fixed point V^*.

D.2 Reward perturbation

If two such MDPs have the same transitions and rewards differing by at most ε_r uniformly, then

$$\|V^* - V'^*\|_\infty \leq \frac{\varepsilon_r}{1-\gamma}.$$

D.3 Transition perturbation

If the transition kernels satisfy

$$\sup_{s,a} \|P(\cdot \mid s,a) - P'(\cdot \mid s,a)\|_{\mathrm{TV}} \leq \varepsilon_P$$

and rewards are bounded by $R_{\max}$, then under the convention

$$|\mathbb{E}_P f - \mathbb{E}_{P'} f| \leq 2\|f\|_\infty \|P - P'\|_{\mathrm{TV}},$$

one obtains

$$\|V^* - V'^*\|_\infty \leq \frac{\varepsilon_r}{1-\gamma} + \frac{2\gamma R_{\max}\varepsilon_P}{(1-\gamma)^2}.$$

D.4 Q-learning

Once the finite MDP is fixed, tabular Q-learning is a separate algorithmic layer. Under finite state and action sets, bounded rewards, discount factor strictly below one, infinite visitation of every state-action pair, and standard stochastic-approximation step sizes, Q-values converge almost surely to the optimal action values. [5] This theorem does not establish that a proposed geometric or categorical model has been correctly reduced to the finite MDP.

Appendix E

Finite Epistemic Refinement and Bisimulation

E.1 Refinement operator

For a finite Kripke frame with finitely many agents, let Π_n be the partition induced by atomic valuations together with successor-block signatures through depth n. Define

$$\Pi_{n+1} = R(\Pi_n),$$

where R splits states with different successor signatures for some agent.

E.2 Finite stopping bound

Every strict refinement increases the number of blocks by at least one. Since there are only $|W|$ states,

$$N_{\text{sat}} \leq |W| - 1.$$

This is a worst-case combinatorial bound rather than a prediction of typical depth.

E.3 Bisimulation at the fixed point

When $\Pi_N = \Pi_{N+1}$, equivalent states have the same atomic valuation and the same sets of successor blocks for every agent. Therefore the equivalence is a bisimulation relation, and the quotient preserves truth of ordinary multi-agent modal formulas.

E.4 Decision-relative refinement

Let U_n be the optimal expected utility available using only the depth-n representation. Then

$$\Delta_n = U_{n+1} - U_n$$

measures marginal decision value. The semantic partition can stabilize independently of this application-specific quantity, so no universal decay law should be inferred without a specified decision model.

Appendix F

Operational Categorical Saturation

F.1 Total boundedness theorem

Let

$$A_0 \subseteq A_1 \subseteq \cdots \subseteq A_\infty \subset (Y, d_Y)$$

be nested observable sets. If A_∞ is totally bounded, then for every $\delta > 0$ there exists N such that

$$A_\infty \subseteq \bigcup_{y \in A_N} B(y, \delta).$$

Proof. Choose a finite $\delta/3$-net F for A_∞. For every net point that is within $\delta/3$ of A_∞, choose a point of A_∞ within $\delta/3$. Only finitely many such points are chosen, so they all occur in one common A_N because the sets are nested. Every point of A_∞ is then within $2\delta/3$ of that A_N point. $\square$

F.2 No automatic physical cutoff

The theorem yields finite effective observational depth at tolerance δ. It does not imply that the underlying abstract categorical structure terminates. A numerical bound on the required depth needs an additional quantitative relation between level and operational distinguishability.

Appendix G

Context Monad Verification

G.1 The Reader construction

For a set of environments G, define

$$T_G(X) = X^G.$$

For $f : X \to Y$,

$$T_G(f)(h) = f \circ h.$$

The unit is

$$\eta_X(x)(g) = x,$$

and multiplication is

$$\mu_X(k)(g) = k(g)(g).$$

Theorem G.1 (Reader monad laws)**.** *The triple* (T_G, η, μ) *is a monad on* **Set**.

Proof. Functoriality follows from composition. For the left unit, if $h : G \to X$ then

$$\mu_X(T_G\eta_X(h))(g) = \eta_X(h(g))(g) = h(g).$$

For the right unit,

$$\mu_X(\eta_{T_G(X)}(h))(g) = h(g).$$

For associativity, if $K : G \to (G \to (G \to X))$, both composites evaluate at g to $K(g)(g)(g)$. Hence the monad laws hold. $\square$

G.2 Context selection

For fixed $g_0 \in G$, the evaluation map

$$\mathrm{ev}_{g_0} : X^G \to X, \qquad h \mapsto h(g_0),$$

is natural in X and satisfies

$$\mathrm{ev}_{g_0} \circ \eta_X = \mathrm{id}_X,$$

and

$$\mathrm{ev}_{g_0} \circ \mu_X = \mathrm{ev}_{g_0} \circ T_G(\mathrm{ev}_{g_0}).$$

Thus it is a valid algebra for the Reader monad.

Appendix H

Boundary Conditions for the Research Programme

The strongest surviving results in this volume are the ones in which the hidden intermediate object can be made explicit. Recursive coordinates become a quotient of a formal vector space. Geometric heuristics become exact one-sided certificates. Infinite mathematical domains admit finite representations once compactness, observables, and tolerance are specified. Geometric learning becomes a finite MDP problem only after a finite approximation and Markov model have actually been constructed. Epistemic decay becomes partition refinement. Physical categorical finiteness becomes an operational distinguishability statement. The context monad becomes a standard environment monad.

These replacements do not weaken the research programme. They identify the exact point at which a further theorem, experiment, or physical model is required.

Bibliography

[1] J. D. Bekenstein, "Universal upper bound on the entropy-to-energy ratio for bounded systems," *Physical Review D* 23 (1981), 287–298, DOI 10.1103/PhysRevD.23.287.

[2] M. Schiffer and J. D. Bekenstein, "Proof of the quantum bound on specific entropy for free fields," Phys. Rev. D 39 (1989), 1109.

[3] R. Longo and F. Xu, "Comment on the Bekenstein bound," arXiv:1802.07184 (2018).

[4] R. Longo, "A Bekenstein-type bound in QFT," arXiv:2409.14408 (2024), giving a rigorous relative-entropy bound for localized states in translation-covariant QFT.

[5] C. J. C. Watkins and P. Dayan, "Q-learning," Machine Learning 8 (1992), 279–292.

[6] N. Haydn, Y. Lacroix, S. Vaienti, "Hitting and return times in ergodic dynamical systems," arXiv:math/0410384.

[7] S. Galatolo, "Hitting time in regular sets and logarithm law for rapidly mixing dynamical systems," arXiv:0906.3416.

[8] T. Bolander and A. Burigana, "Better Bounded Bisimulation Contractions," in Advances in Modal Logic 15 (2024), 249–267.

[9] H. van Ditmarsch, D. Fernández-Duque, W. van der Hoek, "On the definability of simulation and bisimulation in epistemic logic," Journal of Logic and Computation 24(6) (2014), 1209–1227.

[10] G. Lowe, "Breaking and fixing the Needham–Schroeder public-key protocol using FDR," TACAS 1996, LNCS 1055, 147–166.

[11] E. Moggi, "Notions of computation and monads," Information and Computation 93 (1991), 55–92.

[12] S. Mac Lane, *Categories for the Working Mathematician*, 2nd ed., Springer, 1998.

[13] A. Mahajan, "Review of Covering and Packing Numbers," lecture notes; used only for standard covering/packing terminology.

[14] A. W. van der Vaart and J. A. Wellner, *Weak Convergence and Empirical Processes*, Springer, 1996; 2nd ed., 2023, including standard treatments of covering and entropy numbers.

[15] P. B. Pulickal, "The Number of Numbers: The N-Dimensional Sphere Theory," supplied preprint, July 2024.

[16] N. J. Hitchin, "The self-duality equations on a Riemann surface," *Proceedings of the London Mathematical Society* 55 (1987), 59–126, DOI 10.1112/plms/s3-55.1.59.

[17] N. Nitsure, "Moduli space of semistable pairs on a curve," *Proceedings of the London Mathematical Society* 62 (1991), 275–300, DOI 10.1112/plms/s3-62.2.275.

[18] G. Lowe, "An attack on the Needham–Schroeder public-key authentication protocol," *Information Processing Letters* 56 (1995), 131–133, DOI 10.1016/0020-0190(95)00144-2.

[19] The Univalent Foundations Program, *Homotopy Type Theory: Univalent Foundations of Mathematics*, Institute for Advanced Study, 2013.
HoTT Book

[20] G. Brunerie, "Truncations and truncated higher inductive types," Homotopy Type Theory blog, 2012.
HoTT resource

[21] N. Kraus and J. von Raumer, "Coherence via Well-Foundedness: Taming Set-Quotients in Homotopy Type Theory," arXiv:2001.07655, 2020.
arXiv:2001.07655

[22] J. O'Hara, "Pompeiu's theorem and the moduli space of triangles," arXiv:2002.07084, 2020.
arXiv:2002.07084

[23] R. Montgomery, "The three-body problem and the shape sphere," arXiv:1402.0841, 2014.
arXiv:1402.0841

[24] NIST Digital Library of Mathematical Functions, §4.2, "Definitions: Logarithm, Exponential, Powers," current online edition.
`https://dlmf.nist.gov/4.2`

[25] NIST Digital Library of Mathematical Functions, §4.13, "Lambert W-Function," current online edition.
`https://dlmf.nist.gov/4.13`

[26] Bureau International des Poids et Mesures, *The International System of Units (SI Brochure)*, 9th ed.

[27] National Institute of Standards and Technology, "Guide to the SI, Chapter 7: Rules and Style Conventions for Expressing Values of Quantities."
NIST Guide to the SI

What remains genuinely open

Three gaps remain deliberately open rather than being disguised as theorems. First, a physically justified map from a resource budget to a mathematically specified observable probe family is model-dependent; the new resource-to-depth principle is therefore a conjectural rule of thumb whose key missing quantity is a proved per-level distinguishability cost. Second, no general theorem currently connects Hitchin-stack categorical structure to a finite MDP with certified fidelity; the new geometric-to-RL principle is a testable conjectural architecture, and its key missing ingredients are a concrete compact operational sector, approximately Markov-closed finite representation, and explicit transition/action/reward error bounds. Third, the recursive-root formal-coordinate space has no canonical new multiplication beyond its complex quotient without additional algebraic structure. These are not failures of proof technique but missing hypotheses or missing constructions.

Provenance Note

The mathematical reconstruction was developed from a supplied source corpus consisting of studies on recursive-imaginary structures, ellipse and ellipsoid geometry, finite resolution, geometric learning, operational depth, epistemic systems, physical finiteness, and contextual computation. The source works include studies of higher-order imaginary systems, elliptical-cluster anomaly detection, finite numerical representation, Hitchin-stack and learning constructions, operational depth, epistemic decay, physical finiteness, and context monads. Historical formulations are retained where they clarify the mathematical questions treated here. The theorem statements presented in this volume are derived in the stated models or identified with standard results together with their hypotheses. External literature is cited where a claim depends on a standard theorem or where an external mathematical context materially affects the scope of a statement.

Research Notes

For calculations, counterexamples, observations, and future work

This volume examines a collection of mathematical programmes at the boundary between formal structure and operational realization. It develops corrected results for recursive complex-root dynamics, geometric exclusion certificates, finite-resolution representations, epistemic refinement, and contextual computation.

The central strategy is deliberately explicit: define the object, state the hypotheses, identify the exact mathematical consequence, and keep algorithmic, physical, and empirical claims separate. Where an original idea fails, the volume seeks the strongest nearby statement that remains provable.

The resulting programme connects finite-state reasoning, geometric approximation, categorical context, and mathematically testable research questions without requiring a single universal theory.

INDEPENDENT RESEARCH MONOGRAPH

A mathematical notebook of proofs, limits, and open problems